

Aiyagari (1994): Human-Readable Proofs

Exact translation and adequacy audit of the Lean 4 development

Based on i003_replicate_aiyagari

Abstract

This document translates the current Lean 4 formalization into conventional mathematics. Definitions, assumptions, and equilibrium conditions receive the tags (Di), (Ai), and (Ei). Every theorem and lemma is stated with its actual Lean hypotheses and its proof is rewritten for human review. All declarations are accepted by Lean without `sorry`; color instead measures whether the declaration supplies a foundational proof of the intended economic claim. Green means adequate, amber means a rigorous but conditional or narrower result, and red means that the valid Lean statement does not yet establish the economic claim suggested by its name.

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1 Definitions

All scalar quantities are real unless indicated otherwise. A household state is $s = (a, l) \in \mathbb{R}^2$, and a real stochastic process on $(\Omega, \mathcal{F})$ is a map $x : \mathbb{N} \times \Omega \rightarrow \mathbb{R}$. An asset-supply or capital-demand curve is a function $\mathbb{R} \rightarrow \mathbb{R}$.

1.1 Primitives and household accounting

The household primitives are $M = (\beta, r, w, b, l_{\min}, l_{\max}, u, \pi_l)$, where π_l is a measure on labor. Production primitives are $P = (\beta, \delta, l_{\min}, F, F_k, w_K, K)$. Define

$$\lambda(\beta) = \frac{1 - \beta}{\beta}, \quad (\text{D1})$$

$$\phi^{\text{nat}}(w, l_{\min}, r) = \frac{wl_{\min}}{r}, \quad (\text{D2})$$

$$\phi(b, w, l_{\min}, r) = \begin{cases} \min\{b, \phi^{\text{nat}}(w, l_{\min}, r)\}, & r > 0, \\ b, & r \leq 0, \end{cases} \quad (\text{D3})$$

$$\phi^{\text{PV}}(w, l_{\min}, r) = \phi^{\text{nat}}(w, l_{\min}, r), \quad (\text{D4})$$

$$\hat{a} = a + \phi, \quad (\text{D5})$$

$$z(w, l, r, \hat{a}, \phi) = wl + (1 + r)\hat{a} - r\phi, \quad (\text{D6})$$

$$z_{\min}(w, l_{\min}, r, \phi) = wl_{\min} - r\phi, \quad (\text{D7})$$

$$c_A(z) = z - A(z). \quad (\text{D8})$$

For a next-period labor realization l' , resources evolve as

$$T_{M,\phi}(\hat{a}, l') = M.wl' + (1 + M.r)\hat{a} - M.r\phi. \quad (\text{D9})$$

1.2 The Bellman problem

The feasible shifted-asset set at resources z is

$$\Gamma(z) = \{\hat{a} : 0 \leq \hat{a} \leq z\}. \quad (\text{D10})$$

For a candidate value function V , set

$$Q_{M,\phi}V(\hat{a}) = \int V(T_{M,\phi}(\hat{a}, l)) \pi_l(dl), \quad (\text{D11})$$

$$J_{M,\phi,V}(z, \hat{a}) = u(z - \hat{a}) + \beta Q_{M,\phi}V(\hat{a}), \quad (\text{D12})$$

$$(\mathcal{B}_{M,\phi}V)(z) = \sup_{\hat{a} \in \Gamma(z)} J_{M,\phi,V}(z, \hat{a}). \quad (\text{D13})$$

A Bellman fixed point and an optimal policy mean, respectively,

$$\mathcal{B}_{M,\phi}V = V, \quad (\text{D14})$$

$$A(z) \in \Gamma(z), \quad J(z, x) \leq J(z, A(z)) \quad (x \in \Gamma(z)). \quad (\text{D15})$$

1.3 Feasible paths and terminal conditions

A deterministic path (a_t, c_t, l_t) is feasible when

$$c_t \geq 0, \quad c_t + a_{t+1} = wl_t + (1 + r)a_t \quad (t \in \mathbb{N}). \quad (\text{D16})$$

A minimum-income feasible path imposes $l_t = l_{\min}$ in (D16). A stochastic plan is feasible when (D16) holds μ -almost surely at each date:

$$\forall t, \quad c_t(\omega) \geq 0, \quad c_t(\omega) + a_{t+1}(\omega) = wl_t(\omega) + (1 + r)a_t(\omega) \quad \mu\text{-a.s.} \quad (\text{D17})$$

Bounded labor support and adaptedness mean

$$\forall t, \quad l_{\min} \leq l_t(\omega) \leq l_{\max} \quad \mu\text{-a.s.}, \quad (\text{D18})$$

$$x_t : (\Omega, \mathcal{F}_t) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R})) \text{ is measurable for every } t. \quad (\text{D19})$$

The pathwise and almost-sure natural debt restrictions are

$$a_t \geq -\phi^{\text{nat}}(w, l_{\min}, r) \quad \forall t, \quad (\text{D20})$$

$$\forall t, \quad a_t(\omega) \geq -\phi^{\text{nat}}(w, l_{\min}, r) \quad \mu\text{-a.s.} \quad (\text{D21})$$

The Lean no-Ponzi condition requires an actual nonnegative limit:

$$\exists L \geq 0 : \quad \frac{a_{t+1}}{(1+r)^t} \longrightarrow L, \quad (\text{D22})$$

$$\text{condition (D22) holds for } \mu\text{-almost every } \omega. \quad (\text{D23})$$

1.4 Policies, kernels, and aggregation

The auxiliary policy predicates are

$$\beta(1+r)e(z) \leq m(z), \quad (\text{D24})$$

$$z_{\min} < \hat{z}, \quad A(z) = 0 \quad (z \in [z_{\min}, \hat{z}]), \quad (\text{D25})$$

$$A_0(z) \leq A_1(z), \quad c_{A_1}(z) \leq c_{A_0}(z), \quad (\text{D26})$$

$$z^* \leq z \implies T(z, l) \leq z. \quad (\text{D27})$$

If the transition is measurable, the policy-induced Markov kernel is the law of $T_{M,\phi}(A(z), l')$ when $l' \sim \pi_l$:

$$\kappa_A(z, B) = \int \mathbf{1}_B(T_{M,\phi}(A(z), l')) \pi_l(dl'). \quad (\text{D28})$$

Invariance and stationary mean net assets are

$$\mu \kappa_A = \mu, \quad (\text{D29})$$

$$\bar{a}(\phi, A, \mu) = \int (A(z) - \phi) \mu(dz). \quad (\text{D30})$$

For curves $K, E_a : \mathbb{R} \rightarrow \mathbb{R}$, define

$$\text{MC}(K, E_a, r) : \iff K(r) = E_a(r), \quad (\text{D31})$$

$$s(k) = \frac{\delta k}{F(k)}, \quad (\text{D32})$$

$$X(r) = K(r) - E_a(r). \quad (\text{D33})$$

The full-insurance return condition is $r = \lambda(\beta)$. A curve is “nonmonotone” precisely when it is neither monotone nor antitone. Dominance near $\bar{r}$ and one-sided explosion mean

$$\exists \epsilon > 0 : \quad \bar{r} - \epsilon < r < \bar{r} \implies E_a^{\text{certain}}(r) < E_a^{\text{risky}}(r), \quad (\text{D34})$$

$$E_a(r) \rightarrow +\infty \quad (r \uparrow \bar{r}), \quad E_a(r) \rightarrow -\infty \quad (r \downarrow \bar{r}). \quad (\text{D35})$$

A fixed limit is slack when

$$\phi^{\text{nat}}(w, l_{\min}, r) \leq b. \quad (\text{D36})$$

1.5 Growth, debt, and money

The balanced-growth return is

$$r_g^{\text{FI}} = (1 + \lambda(\beta))(1 + g)^\gamma - 1. \quad (\text{D37})$$

Debt-adjusted resources, net assets, and the tax-adjusted floor are

$$R_d(w, l, r, d, a) = wl - rd + (1 + r)a, \quad (\text{D38})$$

$$a^{\text{net}} = a - d, \quad (\text{D39})$$

$$\underline{a}_d = -\frac{wl_{\min} - rd}{r}. \quad (\text{D40})$$

For money m , price p , and a candidate sharp return $r^\#$,

$$a_m = \frac{m - 1}{p}, \quad (\text{D41})$$

$$p^\# = \frac{r^\#}{wl_{\min}}. \quad (\text{D42})$$

Finally, $P.\lambda = \lambda(P.\beta)$ and $P.s(k) = P.\delta k / P.F(k)$.

1.6 Canonical histories and the continuous Bellman state

The refactored household problem uses the canonical history space and product law

$$\Omega = \mathbb{R}^{\mathbb{N}}, \quad \mathbb{P} = \pi_l^{\otimes \mathbb{N}}, \quad l_t(\omega) = \omega_t, \quad \mathcal{F}_t = \sigma(l_0, \dots, l_t). \quad (\text{D43})$$

Resources are measured above their minimum: $x = z - z_{\min} \in \mathbb{R}_+$, and a saving share $s \in [0, 1]$ represents $\hat{a}' = sz$. Thus

$$z = z_{\min} + x, \quad c(x, s) = z(1 - s), \quad \Gamma_s = [0, 1]. \quad (\text{D44})$$

For $l' \in [l_{\min}, l_{\max}]$, next excess resources are

$$x' = \max\{0, (1 + r)sz + w(l' - l_{\min})\}. \quad (\text{D45})$$

In the bounded branch, $C_b(\mathbb{R}_+)$ has the sup norm. For $V \in C_b(\mathbb{R}_+)$, define

$$\begin{aligned} Q_V(x, s) &= u(c(x, s)) + \beta \int V(x'(x, s, l')) \pi_l(dl'), \\ (\mathbb{B}V)(x) &= \max_{s \in [0, 1]} Q_V(x, s). \end{aligned} \quad (\text{D46})$$

The concrete `HouseholdSolution` contains the resulting value and a maximizing policy, together with their defining equations, but contains no uniqueness field.

For the solved bounded household, write $A^*(x)$ for the unique asset policy and define its excess-resource transition by

$$X^*(x, l) = (1 + r)A^*(x) + w(l - l_{\min}). \quad (\text{D47})$$

If x^* is a drift threshold, the compact state and its restricted transition are

$$S_{x^*} = [0, x^*], \quad \hat{X}^*(x, l) = X^*(x, l) \in S_{x^*}. \quad (\text{D48})$$

For a Markov kernel κ , its action on laws and its Cesàro orbit averages are

$$\mathcal{T}_\kappa(\mu) = \kappa \circ_m \mu, \quad (\text{D49})$$

$$\bar{\mu}_n = \frac{1}{n+1} \sum_{i=0}^n \mathcal{T}_\kappa^i(\mu_0). \quad (\text{D50})$$

For a Feller kernel, the dual Markov operator and the oscillation seminorm are

$$(P_\kappa f)(x) = \int f(y) \kappa(x, dy), \quad (\text{D51})$$

$$\text{osc}(f) = \sup_{x,y} |f(x) - f(y)|. \quad (\text{D52})$$

The kernel-level mixing condition used below is the existence of $q \in [0, 1)$ such that

$$\text{osc}(P_\kappa f) \leq q \text{osc}(f) \quad (f \in C_b(S_{x^*})). \quad (\text{D53})$$

The minimum-labor and offset transitions used in the low-income-block argument are

$$T_0(x) = X^*(x, l_{\min}) = (1+r)A^*(x), \quad (\text{D54})$$

$$T_\delta(x) = (1+r)A^*(x) + w\delta, \quad \delta \geq 0. \quad (\text{D55})$$

For an offset history $\boldsymbol{\delta} = (\delta_0, \delta_1, \dots)$, define its finite resource recursion by

$$Z_0(x; \boldsymbol{\delta}) = x, \quad Z_{n+1}(x; \boldsymbol{\delta}) = T_{\delta_n}(Z_n(x; \boldsymbol{\delta})). \quad (\text{D56})$$

For a supported labor history, $\delta_i = l_i - l_{\min}$, so (D56) is exactly the iterated solved transition (D47). The finite-step mixing condition derived below for the household kernel is the weaker requirement that some positive iterate contracts oscillation:

$$\exists m \geq 1, \exists q \in [0, 1) : \quad \text{osc}(P_\kappa^m f) \leq q \text{osc}(f) \quad (f \in C_b(S_{x^*})). \quad (\text{D57})$$

For probability laws μ, ν on $\mathbb{R}$, the refactored convex order and mean-preserving-spread relation are

$$\mu \preceq_{\text{cx}} \nu \iff \begin{cases} \int y d\mu = \int y d\nu, \\ \int f d\mu \leq \int f d\nu \quad \text{for every integrable convex } f. \end{cases} \quad (\text{D58})$$

The shifted-difference form of Miller's convex-marginal continuation class is

$$V \in \mathcal{M} \iff V \text{ is concave and } y \mapsto V(x_1 + y) - V(x_2 + y) \text{ is concave whenever } x_1 \leq x_2. \quad (\text{D59})$$

For a one-step consumption choice, define

$$J_\mu(c; x, V) = u(c) + \beta \int V(R(x - c) + y) \mu(dy), \quad R \geq 0. \quad (\text{D60})$$

2 Assumptions

This section records exactly the structures in **Assumptions.lean**. They are inputs, not conclusions.

The household sign and support assumptions are

$$\begin{aligned} 0 < \beta < 1, \quad r > 0, \quad w > 0, \quad b \geq 0, \quad 0 < l_{\min} < l_{\max}, \\ \pi_l([l_{\min}, l_{\max}]) = 1, \quad \pi_l \text{ is a probability measure.} \end{aligned} \quad (\text{A1})$$

On $[0, \infty)$, utility is continuous, strictly increasing, strictly concave, differentiable on $(0, \infty)$, and

$$u'(c) > 0 \quad (c > 0). \quad (\text{A2})$$

The two boundary regimes are separate:

$$u \text{ is continuous at } 0 \text{ and has a finite right derivative there,} \quad (\text{A3})$$

$$u'(c) \longrightarrow +\infty \quad (c \downarrow 0). \quad (\text{A4})$$

For the bounded finite-at-zero construction, the representative utility is twice continuously differentiable on $(0, \infty)$, has negative second derivative there, and admits finite uniform bounds on $[0, \infty)$:

$$\underline{u} \leq u(c) \leq \bar{u} \quad (c \geq 0), \quad u''(c) < 0 \quad (c > 0). \quad (\text{A5})$$

Borrowing-limit admissibility and endpoint reachability are

$$z_{\min} = wl_{\min} - r\phi \geq 0, \quad (\text{A6})$$

$$\pi_l([l_{\min}, \min\{l_{\max}, l_{\min} + \epsilon\}]) > 0, \quad \pi_l([\max\{l_{\min}, l_{\max} - \epsilon\}, l_{\max}]) > 0 \quad (\epsilon > 0). \quad (\text{A7})$$

The curvature assumptions needed by later layers are convex marginal utility and bounded eventual relative risk aversion:

$$u' \text{ is convex on } (0, \infty), \quad u' \text{ is differentiable on } (c_0, \infty), \quad -\frac{cu''(c)}{u'(c)} \leq \bar{\gamma} \quad (c \geq c_0). \quad (\text{A8})$$

The positive-consumption interface assigns no economic value to $u(0)$. It assumes Inada behavior and the weighted-space growth bound

$$|u(c)| \leq C(1 + c^p), \quad c > 0, \quad C, p \geq 0. \quad (\text{A9})$$

The impatience condition used only by cutoff, Euler, and drift arguments is kept separate from the basic household signs:

$$\beta(1 + r) < 1. \quad (\text{A10})$$

Technology assumptions are

$$0 < P\beta < 1, \quad P\delta \geq 0, \quad P.l_{\min} > 0, \quad K \text{ strictly decreasing,} \quad F(k) > 0 \quad (k > 0). \quad (\text{A11})$$

A resource drift certificate is

$$l_{\min} \leq l_{\max}, \quad z \geq z^*, \quad l \in [l_{\min}, l_{\max}] \implies T(z, l) \leq z. \quad (\text{A12})$$

The only closure input used for differentiating a labor expectation is a local dominated-derivative certificate. For an integrand $F(a, l)$, a candidate derivative $F_a(a, l)$, and a reference action a_0 , it supplies a neighborhood $N \ni a_0$, measurability and integrability, and an integrable envelope g such that

$$\partial_a F(a, l) = F_a(a, l), \quad |F_a(a, l)| \leq g(l) \quad (a \in N, \pi_l\text{-a.e. } l), \quad \int g d\pi_l < \infty. \quad (\text{A13})$$

It does not assume the derivative of the expectation or an Euler equation; both conclusions are derived below. The one remaining global-policy closure used in the Proposition 4 layer is explicitly isolated as **HighResourceConsumptionClosure**:

$$c(x) \longrightarrow +\infty \quad \text{as } x \longrightarrow +\infty. \quad (\text{A14})$$

This is the minimal closure requested for the global passage. It assumes no policy slope, affine policy bound, resource drift, absorbing interval, stationary law, or comparative-static conclusion. The intended later upgrade is to derive (A14) from boundedness and concavity of the solved value and the envelope equation. The former Bellman-contraction, objective-concavity, law-contraction, and invariant-law certificate structures have been removed. Bellman contraction, compact-Feller invariant-law existence, and the generic oscillation-mixing theorem are now proved for concrete mathematical objects. Finite-step splitting is also derived below from the solved household policy, i.i.d. lower-support reachability, and a finite-product Fubini identity. The remaining paper-level closures are the explicitly displayed analytic inputs in (A13)–(A14) and positive minimum-state consumption.

3 Equilibrium

For a measure μ over resources, aggregate labor, assets, and consumption are

$$L(\mu) = \int l d\mu, \quad \mathcal{A}(\phi, A, \mu) = \int (A(z) - \phi) d\mu, \quad C(A, \mu) = \int (z - A(z)) d\mu. \quad (\text{E1})$$

A household solution consists of (V, A) satisfying

$$\mathcal{B}_{M,\phi} V = V, \quad A \text{ satisfies (D15),} \quad (z, l') \mapsto T_{M,\phi}(A(z), l') \text{ is measurable.} \quad (\text{E2})$$

Its kernel is the concrete kernel (D28).

A stationary recursive competitive equilibrium contains $(r, w, k, C, b, \phi, M, V, A, \mu)$ satisfying the primitive matches

$$M.\beta = P.\beta, \quad M.r = r, \quad M.w = w, \quad M.b = b, \quad M.l_{\min} = P.l_{\min}, \quad (\text{E3})$$

the household and stationary conditions

$$(V, A) \text{ satisfies (E2),} \quad \mu \text{ is a probability,} \quad \phi = \phi(b, w, P.l_{\min}, r), \quad \mu \kappa_A = \mu, \quad (\text{E4})$$

firm pricing

$$r = P.F_k(k) - P.\delta, \quad w = P.w_K(k), \quad (\text{E5})$$

and clearing and aggregation

$$L(\pi_l) = 1, \quad \mathcal{A}(\phi, A, \mu) = k, \quad K(r) = k, \quad C(A, \mu) = C, \quad C + P.\delta k = P.F(k). \quad (\text{E6})$$

A full-insurance equilibrium satisfies

$$r = P.\lambda, \quad r = P.F_k(k) - P.\delta, \quad K(r) = k, \quad C + P.\delta k = P.F(k). \quad (\text{E7})$$

A government-debt equilibrium has feasible paths with

$$c_t + a_{t+1} = R_d(w, l_t, r, d, a_t), \quad \mu \text{ probability,} \quad a \text{ integrable,} \quad \int a d\mu = d. \quad (\text{E8})$$

A monetary stationary equilibrium satisfies

$$p > 0, \quad \mu \text{ probability,} \quad a \text{ integrable,} \quad \int a d\mu = 0. \quad (\text{E9})$$

For the complete debt-neutrality construction, translate a shifted net-asset choice $\hat{a}$ into gross government-debt wealth by

$$g_d(\hat{a}) = \hat{a} - \phi + d, \quad \Gamma_d(z) = [d - \phi, z + d - \phi], \quad c_d(z, g) = z + d - \phi - g. \quad (\text{E10})$$

The tax-adjusted transition and Bellman problem are defined directly, rather than by assuming neutrality:

$$T_d(g, l') = R_d(M.w, l', M.r, d, g) - d + \phi, \quad (\text{E11})$$

$$Q_d(V; z, g) = U(c_d(z, g)) + \beta \int V(T_d(g, l')) d\pi_l(l'),$$

$$(\mathcal{B}_d V)(z) = \sup_{g \in \Gamma_d(z)} Q_d(V; z, g). \quad (\text{E12})$$

A complete debt-free stationary equilibrium is a Bellman solution (V, A) and invariant probability μ satisfying

$$\mu \kappa_A = \mu, \quad A - \phi \text{ integrable}, \quad \int (A(z) - \phi) d\mu(z) = 0. \quad (\text{E13})$$

A complete tax-adjusted debt equilibrium requires

$$\mathcal{B}_d V = V, \quad g_d \circ A \text{ maximizes } Q_d, \quad \mu \kappa_A = \mu, \quad g_d \circ A \text{ integrable}, \quad \int g_d(A(z)) d\mu(z) = d. \quad (\text{E14})$$

For the solved bounded household on excess resources, a full-state stationary record $S = (M, \phi, A, \mu)$ requires the primitive packages (A1), (A4), and (A6), probability of μ , invariance under the actual solved kernel, and integrability of net assets. Its constructed mean assets are

$$\mu \kappa_{M, \phi} = \mu, \quad \mathfrak{A}(S) = \int_{\mathbb{R}_+} (A_{M, \phi}(x) - \phi) d\mu(x). \quad (\text{E15})$$

A solved monetary stationary equilibrium for M contains $p > 0$, $M.b = 1/p$, and a stationary record at the effective limit $\phi_m = \phi(1/p, M.w, M.l_{\min}, M.r)$, satisfying

$$\mathfrak{A}(S_m) = 0. \quad (\text{E16})$$

A natural-limit zero-asset equilibrium for $M^\#$ is a stationary record at $\phi^{\text{nat}}(M^\#.w, M^\#.l_{\min}, M^\#.r)$ satisfying

$$\mathfrak{A}(S^\#) = 0. \quad (\text{E17})$$

The primitive-matching relation

`SameHouseholdEnvironmentExceptReturn`

requires two models to have identical $\beta, w, l_{\min}, l_{\max}, U$, and labor law; only their return and borrowing-limit bookkeeping fields may differ.

4 Properties, lemmas, and human-readable proofs

4.1 Immediate equilibrium consequences

Lemma E-L1: `consumption_nonneg`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

If (V, A) is household equilibrium data, then $c_A(z) \geq 0$ for every z . Indeed, optimal-policy feasibility in (D15) gives $A(z) \leq z$, so $c_A(z) = z - A(z) \geq 0$ by (D8).

Lemma E-L2: market_clears**GREEN – Foundationally proved.** **Lean status:** kernel-checked, no `sorryAx`.

In a stationary recursive equilibrium, (E6) gives both $K(r) = k$ and $\mathcal{A}(\phi, A, \mu) = k$. Hence $K(r) = \mathcal{A}(\phi, A, \mu)$, which is market clearing (D31).

4.2 Property 1: debt limits and no Ponzi schemes

Put $d = \phi^{\text{nat}}(w, l_{\min}, r)$ and

$$y_n = \frac{a_{n+1} + d}{(1+r)^n}. \quad (\text{P1.1})$$

Lemma 1.1: minimumIncome_shifted_budget**GREEN – Foundationally proved.** **Lean status:** kernel-checked, no `sorryAx`.

For $r \neq 0$ and a minimum-income feasible path,

$$c_t + (a_{t+1} + d) = (1+r)(a_t + d). \quad (\text{L1.1})$$

By (D2), $rd = wl_{\min}$. Substitute this and the budget equation (D16) with $l_t = l_{\min}$; rearrangement gives (L1.1).

Lemma 1.2: discountedShiftedAssets_antitone**GREEN – Foundationally proved.** **Lean status:** kernel-checked, no `sorryAx`.

If $r > 0$, the sequence (P1.1) is nonincreasing. From (L1.1) and $c_{n+1} \geq 0$, $a_{n+2} + d \leq (1+r)(a_{n+1} + d)$. Division by the positive number $(1+r)^{n+1}$ gives $y_{n+1} \leq y_n$.

Lemma 1.3: tendsto_naturalDebtLimit_div_pow**GREEN – Foundationally proved.** **Lean status:** kernel-checked, no `sorryAx`.

For $r > 0$, $1+r > 1$, and therefore

$$\frac{d}{(1+r)^n} \rightarrow 0. \quad (\text{L1.2})$$

This is the standard geometric-sequence limit multiplied by the constant d .

Property 1: property01_naturalDebt_iff_noPonzi**GREEN – Foundationally proved.** **Lean status:** kernel-checked, no `sorryAx`.

Let $P = (a, c)$ be a feasible predictable plan on the canonical i.i.d. history space. Under bounded labor support and lower-support reachability,

$$[a_t \geq -d \text{ for every } t \text{ a.s.}] \iff \left[\exists L(\omega) \geq 0 : \frac{a_{t+1}(\omega)}{(1+r)^t} \rightarrow L(\omega) \text{ a.s.} \right]. \quad (\text{P1.2})$$

Forward. Iterating and discounting the budget equation bounds the terminal asset by current assets plus the realized present value of bounded labor income. The natural bound controls the terminal remainder, and nonnegative consumption makes the discounted budget sums monotone. Their limit is nonnegative; countable intersection makes the conclusion hold at all dates on one full-measure event.

Reverse. If the natural bound fails at date $s + 1$, predictability gives a positive-measure event, measurable with respect to date- s information, on which $a_{s+1} < -d - \delta$ for some $\delta > 0$. Choose a long future block with every labor draw within ϵ of $l_{\min}$. Independence and lower reachability give a positive-probability intersection. On it, discounted budget summation and no Ponzi imply the opposite lower bound once the low-income block is sufficiently long, a contradiction.

Property 1 (a.s. alias): `property01_naturalDebt_iff_noPonzi_ae`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

This declaration is an explicit alias of (P1.2), retaining the same canonical history, timing, feasibility, support, and reachability assumptions.

Counterexample 1: `natural_bound_does_not_imply_noPonzi_without_feasibility`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Let $a_t = 2^t$ for even t and $a_t = 3 \cdot 2^t$ for odd t . With $w = l_{\min} = r = 1$, $a_t \geq -1$, so (D20) holds. Yet $a_{t+1}/2^t$ alternates between 6 and 2. It is not Cauchy and has no limit, contradicting (D22). Thus feasibility is essential.

Counterexample 2: `noPonzi_does_not_imply_natural_bound_without_feasibility`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Let $a_0 = -2$ and $a_t = 0$ for $t \geq 1$, again with $w = l_{\min} = r = 1$. The discounted future path is identically zero and hence satisfies (D22) with $L = 0$, but $a_0 = -2 < -1$, violating (D20).

4.3 Properties 2–5: dynamic programming and policy shape

Canonical i.i.d. lemmas: `canonicalLabor_map_eq`, `measure_lowerLaborBlock`, `lowerLaborBlock_pos`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Each coordinate projection under (D43) has law π_l . For a finite index set $I = \{t, \dots, t + n\}$ and $B_\epsilon = [l_{\min}, \min\{l_{\max}, l_{\min} + \epsilon\}]$, the product measure cylinder formula gives

$$\mathbb{P}\{l_i \in B_\epsilon \ \forall i \in I\} = \prod_{i \in I} \pi_l(B_\epsilon) > 0. \quad (\text{L2.2})$$

The equality is independence of the canonical coordinates; strict positivity uses lower reachability (A7), factor by factor. This is the probability input used in the reverse direction of Property 1.

Compact maximization lemmas: `exists_compactMaximum_eq`, `compactMaximum_le`, `abs_compactMaximum_sub_le`, `compactMaximum_lipschitz`, `continuous_compactMaximum`, `maximizeObjective_norm_sub_le`, `maximizeObjective_lipschitz`, `continuous_maximizeObjective`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

If f, g are continuous on a nonempty compact action space, both maxima are attained. Evaluating g at a maximizer of f , and conversely, gives

$$|\max f - \max g| \leq \|f - g\|_\infty. \quad (\text{L2.3})$$

Applying this state by state proves that maximization of a bounded continuous state-action objective is nonexpansive and continuous.

Expectation lemmas: `abs_integral_le_uniform_bound`, `abs_integral_bcf_sub_le`, `expectedContinuationBCF_norm_sub_le`, `expectedContinuationBCF_lipschitz`, `continuous_expectedContinuationBCF`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

For a probability measure and bounded continuous V, W ,

$$\left| \int (V - W) d\pi_l \right| \leq \int |V - W| d\pi_l \leq \|V - W\|_\infty. \quad (\text{L2.4})$$

The parametric-integral theorem and continuity of (D45) show that expected continuation is bounded continuous. Taking the supremum proves nonexpansiveness of the continuation operator.

Solved bounded Bellman core: `continuousBellmanOperator_norm_sub_le`, `continuousBellmanOperator_contracting`, `continuous_continuousBellmanOperator_reward_value`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Combining (L2.3)–(L2.4) with (D46) yields directly

$$\|\mathbb{B}V - \mathbb{B}W\|_\infty \leq \beta \|V - W\|_\infty. \quad (\text{L2.5})$$

Since $0 \leq \beta < 1$, this is a contraction on the complete space $C_b(\mathbb{R}_+)$. No contraction certificate is supplied as a hypothesis.

Parameterized Banach fixed points: `parameterizedContractionFixedPoint_isFixedPt`, `dist_parameterizedContractionFixedPoint_le`, `continuous_parameterizedContractionFixedPoint`, `continuous_boundedBellmanValue_reward`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Let $T_\theta : X \rightarrow X$ be a family of maps on a complete metric space, all contracting with the same coefficient $K < 1$, and let V_θ denote the Banach fixed point. Applying the contraction residual estimate for T_θ at the old fixed point V_η gives

$$d(V_\theta, V_\eta) \leq \frac{d(V_\eta, T_\theta V_\eta)}{1 - K}. \quad (\text{L2.5a})$$

If $(\theta, V) \mapsto T_\theta V$ is jointly continuous and $\theta_n \rightarrow \theta$, then the numerator in (L2.5a), with $\eta = \theta$, converges to $d(V_\theta, T_\theta V_\theta) = 0$. The squeeze theorem therefore proves $V_{\theta_n} \rightarrow V_\theta$. This establishes parameter continuity of fixed points without compactness of the value-function space or a supplied fixed-point-continuity field.

For the concrete bounded Bellman operator, the expectation map is nonexpansive, maximization is nonexpansive, and continuous addition and scalar multiplication show that $(R, V) \mapsto \mathbb{B}_R V$ is jointly continuous when the transition and discount factor are fixed. Hence $R \mapsto V_R$ is now proved continuous. Extending this calculation to the price-dependent transition is the remaining model-specific part.

Compact parameterized Bellman and policy theorem:

`continuous_boundedContinuousFamily`, `continuous_compactParameterizedExpectation`,
`continuous_compactParameterizedBellmanOperator`,
`continuous_compactParameterizedBellmanValue`, `continuous_continuousFamilyArgmax`,
`continuous_compactParameterizedSolvedPolicy`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Let Θ be locally compact and metrizable and let the state, action, and shock spaces X, S, L be common compact metric spaces. Suppose the primitive reward $R_\theta \in C_b(X \times S)$ is continuous in the sup norm and the transition $g(\theta, x, s, l)$ is jointly continuous. For fixed V , the Heine–Cantor theorem upgrades joint continuity to uniform continuity over the compact fiber. Probability integration then gives

$$\theta_n \rightarrow \theta \implies \sup_{x,s} \left| \int V(g(\theta_n, x, s, l)) d\pi_l - \int V(g(\theta, x, s, l)) d\pi_l \right| \rightarrow 0. \quad (\text{L2.5b})$$

Together with the one-Lipschitz estimate in V , this proves joint continuity of expected continuation in (θ, V) . Hence

$$(\theta, V) \mapsto \max_{s \in S} \left\{ R_\theta(x, s) + \beta \int V(g(\theta, x, s, l)) d\pi_l \right\} \quad (\text{L2.5c})$$

is jointly continuous as a map into $C_b(X)$ and is a uniform β -contraction in V . Equation (L2.5a) therefore proves sup-norm continuity of the solved value V_θ .

The solved objective is consequently jointly continuous in (θ, x, s) . If primitive strict concavity makes its maximizer unique, every subsequential limit of maximizers remains optimal by continuity and must equal that unique maximizer. Compactness then proves joint continuity of the policy in (θ, x) . This is a proved maximum-theorem conclusion, not a supplied policy-shape or policy-continuity hypothesis.

Scope. This theorem requires one common compact state space that is preserved by every feasible transition in the Bellman problem. Property 6 currently gives an absorbing interval only for the *solved policy*; it does not show that every feasible Bellman action stays in that interval. Thus applying the compact theorem to the untruncated global Aiyagari problem still requires either a uniform truncation-equivalence theorem or a weighted global function space. The generic theorem itself is foundationally proved.

Value and optimizer construction: `boundedBellmanValue_isFixedPoint`,
`boundedBellmanValue_unique`, `boundedAiyagariValue_isFixedPoint`,
`boundedAiyagariValue_unique`, `boundedAiyagariPolicy_isMax`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Banach’s theorem applied to (L2.5) constructs a unique $V^* \in C_b(\mathbb{R}_+)$ satisfying $\mathbb{B}V^* = V^*$. For each x , continuity of $s \mapsto Q_{V^*}(x, s)$ and compactness of $[0, 1]$ produce a maximizing share $s^*(x)$. Hence

$$Q_{V^*}(x, s) \leq Q_{V^*}(x, s^*(x)) \quad (x \geq 0, s \in [0, 1]). \quad (\text{L2.6})$$

The later strict-concavity and maximum-theorem lemmas upgrade this initial selector to the unique continuous shifted-asset policy used in Property 2.

Aiyagari feasibility and transition: `shareConsumption_nonneg`,
`continuous_shareNextExcess`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Under (A6), $z = z_{\min} + x \geq 0$; since $s \in [0, 1]$, (D44) gives $c = z(1 - s) \geq 0$. Formula (D45) is continuous because it is assembled from continuous arithmetic operations and $y \mapsto \max\{0, y\}$; its codomain is $\mathbb{R}_+$ because the maximum is nonnegative.

Lemma 2.1: `beta_mul_one_add_timePreferenceRate`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

For $\beta \neq 0$, direct substitution in (D1) gives

$$\beta(1 + \lambda(\beta)) = 1. \quad (\text{L2.7})$$

Lemma 2.2: `boundedAiyagariValue_unique`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

The concrete bounded Bellman map is a β -contraction in the ordinary sup norm. Thus, if V, W are fixed points,

$$\|V - W\|_{\infty} = \|\mathcal{B}V - \mathcal{B}W\|_{\infty} \leq \beta\|V - W\|_{\infty}.$$

Since $0 < \beta < 1$, the distance is zero and $V = W$. Banach's theorem also constructs the fixed point from the primitive Bellman operator.

Lemma 2.3: `boundedOptimalAsset_unique`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Bellman iteration preserves concavity, so the solved continuation value is concave. Adding strictly concave current utility makes the objective strictly concave in shifted assets on $[0, z]$. Two distinct maximizers would violate the defining strict-concavity inequality at their midpoint; hence the optimizer is unique.

Envelope theorem: `boundedAiyagari_envelope_of_consumption_pos`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Fix $x > 0$, write $a = A(x)$, $z = z_{\min} + x$, and suppose $c = z - a > 0$. Holding a fixed as current resources vary gives

$$s(y) = u(z_{\min} + y - a) + \beta G(a), \quad G(a) = \int V((1 + r)a + w(l - l_{\min})) d\pi_l.$$

This action remains feasible for y near x , hence $s(y) \leq V(y)$, while Bellman optimality gives $s(x) = V(x)$. Concavity supplies both one-sided derivatives of V . The touching lower support and the ordering of concave one-sided slopes force both to equal $s'(x) = u'(c)$. Thus

$$V'(x) = u'(c(x)). \quad (\text{P2.1})$$

No derivative of the policy is used or assumed.

Euler system: boundedAiyagari_euler_inequality**GREEN – Foundationally proved.** **Lean status:** kernel-checked, no `sorryAx`.

Under (A13), dominated differentiation first proves

$$G'(a) = \int (1+r)V'((1+r)a + w(l - l_{\min})) d\pi_l.$$

The objective derivative is $-u'(z - a) + \beta G'(a)$. At the lower asset boundary, every sufficiently small rightward deviation is feasible and cannot improve the objective, so this derivative is nonpositive. If $a > 0$, positive consumption also gives $a < z$, making the optimizer interior; Fermat's theorem then makes the derivative zero. Therefore

$$u'(c(x)) \geq \beta(1+r) \int V'(x'(x, l)) d\pi_l, \quad \text{with equality if } A(x) > 0. \quad (\text{P2.2})$$

Property 2: property02_value_and_policy_regularity**AMBER – Formally proved but incomplete.** **Lean status:** kernel-checked, no `sorryAx`.

Under the bounded finite-at-zero assumptions, Lean constructs `solveBoundedHousehold`, denoted S , from primitives alone. The theorem proves: $S.V$ is the unique bounded continuous fixed point; $S.V$ is concave; each feasible objective is strictly concave; $S.A$ is feasible, optimal, unique, and continuous. Policy continuity is proved by the maximum theorem: every convergent state sequence has a convergent subsequence of saving shares, continuity passes optimality to the limit, and optimizer uniqueness identifies that limit with $S.A(x)$. The numbered declaration now also exposes the envelope identity (P2.1) and, whenever the explicit local domination certificate (A13) is supplied, the Euler inequality and complementary-slackness equality (P2.2).

Missing foundational step. Prove the domination certificate globally from primitive regularity. For parameter continuity, either prove a common compact truncation is exactly equivalent to the unbounded household problem on admissible parameter sets, then apply (L2.5b)–(L2.5c), or construct the global weighted function space directly. Finally specialize the now-proved compact unique argmax theorem to the household objective and construct the distinct positive-consumption weighted Inada theory.

Property 3: property03_binding_region_at_low_resources**AMBER – Formally proved but incomplete.** **Lean status:** kernel-checked, no `sorryAx`.

Assume impatience $\beta(1+r) < 1$, positive optimal consumption at the minimum excess-resource state, and the local dominated-differentiation condition (A13) at every policy choice. Put $m(x) = u'(c(x))$. The envelope and Euler equations imply that whenever $A(x) > 0$,

$$m(x) = \beta(1+r) \int m(x'(A(x), l)) d\pi_l(l) \leq \beta(1+r)m(0). \quad (\text{L3.1})$$

Indeed, $A(x) > 0$ makes every next state positive; monotonicity of consumption and concavity of utility give $m(x'(A(x), l)) \leq m(0)$. At $x = 0$, (L3.1) and $\beta(1+r) < 1$ contradict $m(0) > 0$, proving $A(0) = 0$.

Continuity of consumption and of marginal utility at the positive number $c(0)$ yields a $\delta > 0$ such that, for $0 \leq x < \delta$,

$$\beta(1+r)m(0) < m(x).$$

If saving were positive at such an x , (L3.1) would give the reverse weak inequality. Taking $\hat{x} = \delta/2$, the Lean theorem therefore constructs a genuinely nondegenerate cutoff:

$$\hat{x} > 0, \quad 0 \leq x \leq \hat{x} \implies A(x) = 0 \quad \text{and} \quad c(x) = z_{\min} + x. \quad (\text{P3.1})$$

The supporting declarations are `boundedAiyagariAssetPolicy_zero_at_minimum`, `boundedAiyagari_marginal_le_minimum_of_policy_pos`, and `boundedAiyagari_binding_cutoff`.

Missing foundational step. Derive positive minimum-state consumption and the global family of dominated-differentiation certificates from the primitive finite-at-zero utility and labor assumptions, rather than supplying them to the numbered theorem.

Inada claim: `property03_inada_exception`

RED – Intended claim not yet established. **Lean status:** kernel-checked, no `sorryAx`. The Lean statement assumes $A(z) > 0$ for every $z > z_{\min}$ and concludes $A(z) \neq 0$. The proof is the order fact “strictly positive implies nonzero.”

Missing foundational step. Use (A4), feasibility, and optimality to prove policy positivity. The current theorem neither mentions nor uses the Inada assumption.

Property 4: `property04_monotonicity_and_one_for_one_response`

AMBER – Formally proved but incomplete. **Lean status:** kernel-checked, no `sorryAx`. For the actual solved bounded household, Lean now proves that both assets and consumption are weakly increasing and continuous. Revealed preference and strict concavity rule out a decrease in assets. A second revealed-preference argument exchanges the two consumption choices; strict optimality then contradicts concavity of expected continuation value if consumption falls. Consequently, for $x_1 \leq x_2$,

$$0 \leq A(x_2) - A(x_1) \leq x_2 - x_1. \quad (\text{P4.1})$$

Thus the weak derivative bounds $0 \leq A' \leq 1$ follow at every point where the policy is differentiable, without assuming them as policy-shape inputs.

Missing foundational step. Use the envelope/Euler system plus explicit smoothness and nondegeneracy to prove strict monotonicity and $0 < A'(z) < 1$ off the binding region; combine this with Property 3 for the one-for-one binding response.

Lemma 5.1: `consumptionShiftedDown_of_policyShiftedUp`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`. If $A_0(z) \leq A_1(z)$, then by (D8) $c_{A_1}(z) = z - A_1(z) \leq z - A_0(z) = c_{A_0}(z)$.

Same-law aggregation: `stationaryMeanAssets_mono_same_law`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`. If both policies are integrated against the *same* probability law μ and $A_0(z) \leq A_1(z)$ pointwise, integrability and monotonicity of the integral give

$$\int (A_0(z) - \phi) d\mu(z) \leq \int (A_1(z) - \phi) d\mu(z).$$

This lemma deliberately fixes the law; it does not compare the two endogenous invariant laws generated by different shock distributions.

Convex-order and saving-gap comparison: `integral_mono_of_convexOrder_concave`,
`continuationSavingGap_mono_of_meanPreservingSpread`,
`oneStepObjectiveSavingGap_mono_of_meanPreservingSpread`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

If $\mu \preceq_{\text{cx}} \nu$, applying (D58) to the negative of an integrable concave function gives

$$f \text{ concave} \implies \int f d\nu \leq \int f d\mu. \quad (\text{L5.1})$$

Let $c_L \leq c_H$. Since $R(x - c_H) \leq R(x - c_L)$ and $V \in \mathcal{M}$,

$$y \mapsto V(R(x - c_H) + y) - V(R(x - c_L) + y)$$

is concave. Applying (L5.1) and rearranging proves Miller’s four-point comparison

$$\begin{aligned} & \int V(R(x - c_L) + y) d\mu - \int V(R(x - c_H) + y) d\mu \\ & \leq \int V(R(x - c_L) + y) d\nu - \int V(R(x - c_H) + y) d\nu. \end{aligned} \quad (\text{L5.2})$$

Multiplication by $\beta \geq 0$ and cancellation of current utility gives the same inequality for $J_\mu(c_L) - J_\mu(c_H)$ and $J_\nu(c_L) - J_\nu(c_H)$.

One-step optimizer comparison:

`optimalConsumption_antitone_of_meanPreservingSpread_sameContinuation`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Let c_0 uniquely maximize J_μ and c_1 uniquely maximize J_ν , with the same continuation function $V \in \mathcal{M}$. If $c_0 < c_1$, baseline optimality makes $J_\mu(c_0) - J_\mu(c_1) \geq 0$. Equation (L5.2) then gives $J_\nu(c_0) - J_\nu(c_1) \geq 0$. Risky-law optimality gives the reverse inequality, so both choices maximize J_ν ; uniqueness forces $c_0 = c_1$, a contradiction. Therefore

$$c_1 \leq c_0, \quad x - c_0 \leq x - c_1. \quad (\text{L5.3})$$

Property 5: `property05_mean_preserving_spread`

AMBER – Formally proved but incomplete. **Lean status:** kernel-checked, no `sorryAx`.

The numbered theorem now states (L5.3) for arbitrary continuous shock laws related by the genuine convex order (D58); it no longer uses arbitrary policies or unrelated Dirac laws. Thus the one-step Sibley–Miller single-crossing core is foundationally proved.

Missing foundational step. In the infinite-horizon household, the two shock laws generate different fixed-point continuation values V_μ and V_ν . Complete Miller’s Lemmas 1–3 in the present continuous-state Bellman space: prove that the Bellman operator preserves (D59), that its fixed point remains in this class, and that $V_\nu - V_\mu$ is nondecreasing. This will let (L5.3) compare the two solved policies rather than two objectives with a common continuation value. Aggregate stationary assets must then be handled under the two endogenous invariant laws; pointwise ordering alone

does not order those two integrals.

4.4 Properties 6–10: stationary dynamics and asset supply

RRA growth lemmas: `EventualRelativeRiskAversionBound.marginal_ratio_le_rpow`,
`EventualRelativeRiskAversionBound.tendsto_marginal_ratio_add_const`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

For $c_0 \leq a \leq b$, differentiate $g(c) = \log u'(c) + \bar{\gamma} \log c$. Assumption (A8) gives

$$g'(c) = \frac{u''(c)}{u'(c)} + \frac{\bar{\gamma}}{c} \geq 0.$$

The mean-value monotonicity theorem and exponentiation therefore yield

$$\frac{u'(a)}{u'(b)} \leq \left(\frac{b}{a}\right)^{\bar{\gamma}}. \quad (\text{L6.1})$$

For each fixed $d \geq 0$, decreasing marginal utility supplies the lower bound $1 \leq u'(c)/u'(c+d)$, while (L6.1) supplies the upper bound $(1 + d/c)^{\bar{\gamma}}$. Since the latter converges to one,

$$\lim_{c \rightarrow \infty} \frac{u'(c)}{u'(c+d)} = 1. \quad (\text{L6.2})$$

These are the relative-risk-aversion steps in Aiyagari’s 1993 Proposition 4; no drift or affine policy bound is assumed in either lemma.

Bounded labor-gap comparison: `rra_marginal_ratio_bounded_gap_tendsto_one`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Let c be increasing, $x_-(x) \leq x_+(x)$, and suppose $c(x_+(x)) - c(x_-(x)) \leq d$ for a fixed $d \geq 0$. If $c(x_-(x)) \rightarrow \infty$, marginal-utility monotonicity and (L6.2) give

$$1 \leq \frac{u'(c(x_-(x)))}{u'(c(x_+(x)))} \leq \frac{u'(c(x_-(x)))}{u'(c(x_-(x)) + d)} \rightarrow 1. \quad (\text{L6.3})$$

The squeeze theorem proves that the middle ratio converges to one.

Proposition 4, unbounded-policy branch:

`bounded_resource_dynamics_of_policy_tendsto_atTop`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Write $R = 1 + r$, let A and c be the solved Bellman policy and its induced consumption, and assume $A(x) \rightarrow \infty$. The lowest and highest possible next excess-resource states are

$$x_-(x) = RA(x), \quad x_+(x) = RA(x) + w(l_{\max} - l_{\min}). \quad (\text{P6.1})$$

Policy and consumption monotonicity imply $x_-(x) \leq x_+(x)$ and $0 \leq c(x_+(x)) - c(x_-(x)) \leq w(l_{\max} - l_{\min})$. By (A14), $c(x_-(x)) \rightarrow \infty$; hence (L6.3) applies.

Choose $\varepsilon > 0$ so that

$$\beta R(1 + \varepsilon) < 1. \quad (\text{P6.2})$$

For all sufficiently large x , (L6.3) yields $u'(c(x_-)) < (1 + \varepsilon)u'(c(x_+))$. Every labor realization produces $X' = RA(x) + w(l - l_{\min}) \in [x_-, x_+]$. The envelope equation and decreasing marginal utility therefore imply

$$V'(X') = u'(c(X')) \leq u'(c(x_-)) < (1 + \varepsilon)u'(c(x_+)).$$

Integrating this inequality and using the interior Euler equality gives

$$V'(x) = u'(c(x)) \leq \beta R(1 + \varepsilon)u'(c(x_+)) < V'(x_+). \quad (\text{P6.3})$$

If $x < x_+$, concavity of V instead forces $V'(x_+) \leq [V(x_+) - V(x)]/(x_+ - x) \leq V'(x)$, contradicting (P6.3). Thus $x_+(x) \leq x$ for all sufficiently large x , and so every feasible next state satisfies $X' \leq x$.

Property 6: `property06_bounded_resource_dynamics`

AMBER – Formally proved but incomplete. **Lean status:** kernel-checked, no `sorryAx`. The theorem divides into exhaustive cases for the solved policy. If A is bounded above by K , then

$$X' \leq RK + w(l_{\max} - l_{\min}),$$

so any larger excess-resource state is a drift threshold. If A is not bounded above, its proved monotonicity implies $A(x) \rightarrow \infty$, and the preceding unbounded-policy theorem applies. Consequently

$$\exists x^* < \infty \forall x \geq x^* \forall l \in [l_{\min}, l_{\max}], \quad X'(x, l) \leq x. \quad (\text{P6.4})$$

Missing foundational step. The drift conclusion is now proved from the solved household policy, the Euler and envelope equations, bounded shocks, impatience, and the primitive RRA bound. To upgrade the paper-level status to green, derive the sole remaining global closure (A14) from the bounded Bellman solution rather than assuming it.

Feller and monotonicity foundations: `iidTransitionFeller_of_continuous`, `feller_solvedResourceTransition`, `monotone_solvedResourceTransition`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`. The solved transition on excess resources is

$$X'(x, l') = (1 + r)A(x) + w(l' - l_{\min}).$$

Policy continuity and ordinary algebra make $(x, l') \mapsto X'(x, l')$ jointly continuous. For every bounded continuous test function f , the integrand $f(X'(x, l'))$ is jointly continuous and uniformly bounded on the compact labor support. Continuity of parametric integration therefore gives $x \mapsto \int f(X'(x, l')) \pi_l(dl')$ continuous: the household kernel is Feller. Policy monotonicity and $1 + r > 0$ also give $x_1 \leq x_2 \Rightarrow X'(x_1, l') \leq X'(x_2, l')$ for every shock.

Absorbing compact household kernel: `solvedNextExcess_le_driftThreshold`, `feller_absorbingSolvedKernel`, `exists_invariantProbability_absorbingSolvedKernel`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Let x^* satisfy the Property 6 drift inequality. If $x \leq x^*$, monotonicity of (D47) gives

$$X^*(x, l) \leq X^*(x^*, l) \leq x^*.$$

Thus S_{x^*} in (D48) is positively invariant. Restricting the jointly continuous transition to this interval preserves joint continuity, and hence the induced kernel on S_{x^*} is Feller.

Krylov–Bogoliubov existence: cesaro_residual_tendsto_zero_of_map_average, exists_fixedPoint_of_compact_cesaro, continuous_markovLawEvolution_of_feller, markovLawEvolution_probabilityCesaroAverage, integral_iterate_markovLawEvolution, exists_invariantProbability_of_feller_compact, solvedHousehold_invariantProbability_exists

GREEN – Foundationally proved. **Lean status:** kernel-checked, no sorryAx.

Kernel linearity maps (D50) into the shifted orbit average. Therefore, for every bounded continuous f , telescoping gives

$$\left| \int f d\mathcal{T}_\kappa(\bar{\mu}_n) - \int f d\bar{\mu}_n \right| \leq \frac{2\|f\|_\infty}{n+1} \longrightarrow 0. \quad (\text{L7.1})$$

The space of probability laws on compact S_{x^*} is compact. Hence a subsequence $\bar{\mu}_{n_j}$ converges weakly to some μ^* . Feller continuity makes $\mathcal{T}_\kappa$ continuous, while (L7.1) forces both $\mathcal{T}_\kappa(\bar{\mu}_{n_j})$ and $\bar{\mu}_{n_j}$ to have the same limit. Uniqueness of weak limits then yields $\mathcal{T}_\kappa(\mu^*) = \mu^*$. Combining this theorem with Property 6 proves existence for the actual solved household kernel, without assuming an invariant law or a contraction of laws.

Oscillation mixing, uniqueness, and stability: bcfOscillation_iterate_le, bcfOscillation_fellerMarkovOperator_le_two, bcfOscillation_mul_iterate_le, exists_unique_invariant_and_tendsto_of_feller_mixing

GREEN – Foundationally proved. **Lean status:** kernel-checked, no sorryAx.

Under (D53), induction gives

$$\text{osc}(P_\kappa^n f) \leq q^n \text{osc}(f). \quad (\text{L7.2})$$

For any probability laws μ, ν , centering f at an arbitrary state and applying the sup-norm integral bound gives

$$\left| \int f d\mu - \int f d\nu \right| \leq 2 \text{osc}(f). \quad (\text{L7.3})$$

Duality of (D49) and (D51) yields $\int f d\mathcal{T}_\kappa^n \mu = \int P_\kappa^n f d\mu$. Applying (L7.3) to $P_\kappa^n f$, then (L7.2), proves

$$\left| \int f d\mathcal{T}_\kappa^n \mu_0 - \int f d\mu^* \right| \leq 2q^n \text{osc}(f) \longrightarrow 0. \quad (\text{L7.4})$$

Thus every initial law converges weakly to an invariant law. Applying the same result to a second invariant law makes a constant sequence converge to μ^* , proving uniqueness.

Finite-step mixing controls the original chain:

`tendsto_bcfOscillation_iterate_of_finiteStepMixing,`
`tendsto_iterate_of_kernelIterateOscillationMixing,`
`exists_unique_invariant_and_tendsto_of_feller_finiteStepMixing`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Assume (D57). Write $n = k_n m + j_n$, with $k_n = \lfloor n/m \rfloor$ and $0 \leq j_n < m$. A general Markov operator expands oscillation by at most a factor two, whereas each complete block contracts it by q . Hence

$$\text{osc}(P_\kappa^n f) \leq 2^{j_n} q^{k_n} \text{osc}(f) \leq 2^m q^{\lfloor n/m \rfloor} \text{osc}(f) \longrightarrow 0. \quad (\text{L7.9})$$

Substituting (L7.9) for (L7.2) in the duality argument (L7.3)–(L7.4) proves convergence at every original date, not only along multiples of m . Invariance then proves uniqueness exactly as before. Thus no separate skeleton-to-original-chain assumption is needed.

Common-shock splitting implies mixing:

`kernelOscillationMixing_of_oneStepSplitting`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Suppose a measurable shock event S has probability $p > 0$ and, on S , the transition is independent of the current state. For any x, y , the integrands defining $P_\kappa f(x)$ and $P_\kappa f(y)$ coincide on S . On S^c their absolute difference is at most $\text{osc}(f)$. Hence

$$|P_\kappa f(x) - P_\kappa f(y)| \leq (1 - p) \text{osc}(f), \quad (\text{L7.5})$$

so (D53) holds with $q = 1 - p < 1$.

Minimum-income strict descent: `minimumLaborTransition_strictly_decreases`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Assume $x > 0$. If $A^*(x) = 0$, then (D54) gives $T_0(x) = 0 < x$. Otherwise the asset choice is interior and the Euler equality applies. Suppose toward a contradiction that $T_0(x) \geq x$. Since every supported labor realization satisfies $l' \geq l_{\min}$, $X^*(x, l') \geq T_0(x) \geq x$. Monotonicity of solved consumption and decreasing marginal utility therefore imply, for every l' , that

$$V'(X^*(x, l')) = u'(c^*(X^*(x, l'))) \leq u'(c^*(x)).$$

Substitution into the interior Euler equation yields

$$u'(c^*(x)) \leq \beta(1 + r)u'(c^*(x)) < u'(c^*(x)), \quad (\text{L7.6})$$

where the final inequality uses (A9) and positive marginal utility. This contradiction proves $T_0(x) < x$ at every $x > 0$.

Uniform finite entrance under minimum labor:

`tendsto_iterate_zero_of_monotone_strictDecrease,`
`exists_uniform_iterate_lt_of_monotone_strictDecrease,`
`minimumLaborBlock_uniformly_enters_bindingRegion`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

The map T_0 is continuous and increasing, fixes zero, and is strictly below the identity on $(0, \infty)$ by (L7.6). Hence the sequence $T_0^n(x^*)$ is decreasing and has a limit $\bar{x} \geq 0$. Continuity gives $T_0(\bar{x}) = \bar{x}$; strict descent rules out $\bar{x} > 0$, so

$$T_0^n(x^*) \longrightarrow 0. \quad (\text{L7.7})$$

Let $\hat{x} > 0$ be the binding cutoff from Property 3. Equation (L7.7) supplies an N with $T_0^N(x^*) < \hat{x}$. Monotonicity of T_0^N then gives $T_0^N(x) < \hat{x}$ simultaneously for every $x \in [0, x^*]$.

Positive-probability low-income entrance block:

`exists_positive_lowOffset_uniformly_enters_bindingRegion,`
`lowOffsetBlockTransition_le_constant,`
`lowLaborNeighborhoodBlock_uniformly_enters_bindingRegion,`
`lowIncomeBlock_positive_and_uniformEntrance`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

For fixed N , continuity of (D55) and finite composition imply that $\delta \mapsto T_\delta^N(x^*)$ is continuous. The strict inequality in (L7.7) therefore persists for some $\varepsilon > 0$:

$$T_\varepsilon^N(x^*) < \hat{x}. \quad (\text{L7.8})$$

Both T_δ and the policy are increasing in their arguments. Induction in (D56) thus shows that, whenever $x \leq x^*$ and $0 \leq \delta_i \leq \varepsilon$ for $i < N$,

$$Z_N(x; \delta) \leq T_\varepsilon^N(x^*) < \hat{x}.$$

Finally, lower-support reachability (A4) and independence give the canonical event $\{l_i \leq l_{\min} + \varepsilon : 0 \leq i < N\}$ strictly positive probability. Thus the uniform low-income block needed by Aiyagari is now a proved consequence of the solved policy, the Euler equation, and primitive labor reachability; it is not a recurrence hypothesis.

Actual household block has a common reset: `supportedLowerLaborBlock_pos,`
`continuous_absorbingSolvedBlockTransition,`
`exists_oneStepSplitting_absorbingSolvedBlockTransition,`
`kernelIterateOscillationMixing_of_blockSplitting`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Put the supported labor law on the countable product history space and iterate the actual absorbing random map for $N + 1$ dates. The first N coordinates form the proved positive-probability lower block. They send every initial $x \in S_{x^*}$ below $\hat{x}$, where $A^* = 0$. Consequently, for the additional shock l_N ,

$$X_{N+1}(x; l_0, \dots, l_N) = w(l_N - l_{\min}) \quad \text{for every } x \in S_{x^*}, \quad (\text{L7.10})$$

so the concrete finite-history transition satisfies the common-reset splitting structure. Its joint continuity is proved recursively.

Finite-product representation and derived household mixing:

```
finiteIIDBlockTransition_succ_last, measurable_finiteIIDBlockTransition,
continuous_finiteIIDBlockTransition,
integral_finiteIIDBlockTransition_eq_operator_iterate,
fellerMarkovOperator_finiteIIDBlock_eq_iterate,
finiteSupportedLowerLaborBlock_pos,
exists_oneStepSplitting_finite_absorbingSolvedBlockTransition,
absorbingSolvedKernel_finiteStepMixing
```

GREEN – Foundationally proved. Lean status: kernel-checked, no sorryAx.

Let $\pi_l^{\otimes m}$ be the finite product law and let $\Phi_m(x; l_0, \dots, l_{m-1})$ denote chronological iteration of the actual one-period random map. Induction on m , using finite-product Fubini at the successor step, proves the exact operator identity

$$\int f(\Phi_m(x; \mathbf{l})) d\pi_l^{\otimes m}(\mathbf{l}) = P_\kappa^m f(x). \quad (\text{L7.11})$$

The first N coordinates of the concrete finite product are restricted to the positive-probability lower block, while the last coordinate is free. Equation (L7.10) makes Φ_{N+1} a common reset on this event. Applying (L7.5) to the block kernel and then substituting (L7.11) yields (D57) for the actual one-period household kernel. Thus finite-step mixing is a theorem, not a hypothesis of Property 7.

Property 7: property07_unique_stable_invariant_distribution

AMBER – Formally proved but incomplete. Lean status: kernel-checked, no sorryAx.

Property 6 constructs x^* ; the preceding compact Feller theorem then constructs an invariant probability on S_{x^*} . Lower-support reachability, the low-income reset block, and (L7.11) prove the finite-step oscillation condition (D57) for the actual restricted household kernel. The generic mixing theorem therefore gives

$$\mathcal{T}_\kappa(\mu^*) = \mu^*, \quad \mu^* \text{ is unique}, \quad \mathcal{T}_\kappa^n(\mu_0) \longrightarrow \mu^* \quad \forall \mu_0. \quad (\text{P7.1})$$

Missing foundational step. The stationarity and mixing argument itself is now foundationally proved. The numbered theorem remains amber at the paper level because it still takes positive minimum-state consumption, local dominated differentiation (A13), and the Property 6 high-resource closure (A14) as explicit analytic inputs. Deriving those inputs from the primitive utility and solved Bellman problem would upgrade the complete claim.

```
Continuity of unique invariant laws: continuous_integral_bcf_probabilityMeasure,
continuous_fixedPoint_of_compact_unique,
continuous_parameterizedMarkovLawEvolution_of_continuous_fellerOperator,
continuous_invariantProbability_of_jointlyContinuous_evolution
```

GREEN – Foundationally proved. Lean status: kernel-checked, no sorryAx.

Let Θ be a sequential parameter space, let $\mathcal{P}(S)$ be the compact Hausdorff space of probability measures on a compact metric state space, and let $\mathcal{T} : \Theta \times \mathcal{P}(S) \rightarrow \mathcal{P}(S)$ be jointly continuous. Suppose that for every θ there is a unique fixed point μ_θ :

$$\mathcal{T}(\theta, \mu_\theta) = \mu_\theta, \quad \mathcal{T}(\theta, \nu) = \nu \implies \nu = \mu_\theta. \quad (\text{L7.12})$$

Then $\theta \mapsto \mu_\theta$ is continuous. Indeed, take $\theta_n \rightarrow \theta$. Compactness makes every cluster subsequence of μ_{θ_n} converge further to some $\bar{\mu}$. Joint continuity and (L7.12) imply $\mathcal{T}(\theta, \bar{\mu}) = \bar{\mu}$; uniqueness gives $\bar{\mu} = \mu_\theta$. Thus every cluster point is μ_θ , and compactness forces the whole sequence to converge. Lean proves this first for arbitrary compact fixed-point spaces and then specializes it to Markov law evolution. No continuity of the invariant-law selection is assumed.

The required joint continuity of law evolution is itself reduced to a sup-norm statement about Feller operators. For bounded continuous f, g and probability laws μ, ν ,

$$\left| \int g d\nu - \int f d\mu \right| \leq \|g - f\|_\infty + \left| \int f d\nu - \int f d\mu \right|. \quad (\text{L7.13})$$

Thus joint integration is continuous in the sup-norm integrand and weak law. Consequently, if $\theta \mapsto P_\theta f$ is sup-norm continuous for every bounded continuous f , then $(\theta, \mu) \mapsto \mathcal{T}_\theta(\mu)$ is jointly continuous. The only remaining model-specific continuity step is therefore continuity of the solved household Feller operator, not an invariant-measure assumption.

Remaining household application. To obtain continuity in (b, w, r) , construct a common compact absorbing interval on each admissible compact parameter set and prove joint continuity of the solved household law evolution there. The present primitives bundle prices into a single model, so that parameterized-policy theorem is not yet available.

For n states with fixed weights ω_i , define

$$E_a(r) = \sum_{i=1}^n \omega_i (A_i(r) - \phi(r)). \quad (\text{P8.1})$$

Lemma 8.1: finiteStateMeanAssets_continuous

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

If ϕ and every A_i are continuous, each summand in (P8.1) is continuous. A finite sum of continuous functions is continuous.

Witness 8.2: continuous_nonmonotone_finiteState_supply

RED – Intended claim not yet established. **Lean status:** kernel-checked, no `sorryAx`.

Take one state, $\omega_1 = 1$, $\phi = 0$, and $A_1(r) = r^2$. Then $E_a(r) = r^2$, which is continuous, decreases from -1 to 0 , and increases from 0 to 1 ; hence it is neither monotone nor antitone.

Missing foundational step. Show that this policy is optimal in an admissible finite-state Aiyagari household model, or replace it with a certified CRRA witness.

Property 8: property08_continuity_and_possible_nonmonotonicity

RED – Intended claim not yet established. **Lean status:** kernel-checked, no `sorryAx`.

Continuity of (P8.1) follows from Lemma 8.1, and the existential nonmonotone component is supplied by Witness 8.2.

Missing foundational step. The continuity portion is rigorous, but the nonmonotonicity witness must be an optimal policy and invariant distribution from an admissible Aiyagari model.

Lemma 9.1: reciprocal_gap_diverges

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

As $r \uparrow \lambda$, $\lambda - r \downarrow 0$ through positive values. Continuity of subtraction and inversion therefore gives

$$(\lambda - r)^{-1} \longrightarrow +\infty. \quad (\text{L9.1})$$

Property 9: property09_explosion_at_time_preference_rate

AMBER – Formally proved but incomplete. **Lean status:** kernel-checked, no `sorryAx`.

If eventually, as $r \uparrow \lambda$,

$$(\lambda - r)^{-1} \leq E_a(r), \quad (\text{P9.1})$$

then Lemma 9.1 and order comparison of limits imply $E_a(r) \rightarrow +\infty$, namely (D35).

Missing foundational step. Derive (P9.1) from optimal saving, discounting, bounded shocks, and risk-aversion conditions.

Property 10: property10_uncertainty_raises_assets_near_benchmark

AMBER – Formally proved but incomplete. **Lean status:** kernel-checked, no `sorryAx`.

Let risky and certain asset supply be continuous at λ , with $E_a^c(\lambda) < E_a^r(\lambda)$. Strict inequality is open, so it persists on some interval (l, h) containing λ . Choosing $\epsilon = \lambda - l > 0$ yields (D34).

Missing foundational step. Derive the strict gap at λ from a mean-preserving spread and the associated invariant distributions.

4.5 Properties 11–17: general equilibrium

Property 11: property11_factor_price_and_capital_signs

AMBER – Formally proved but incomplete. **Lean status:** kernel-checked, no `sorryAx`.

Let $X = K - E_a^{IM}$. Clearing at r^{IM} gives $X(r^{IM}) = 0$, while excess supply at r^{FI} gives $X(r^{FI}) < 0$. If X is strictly decreasing, these facts force $r^{IM} < r^{FI}$. Equation (E7) gives $r^{FI} = P.\lambda$. Since capital demand is strictly decreasing by (A11),

$$r^{IM} < r^{FI} = P.\lambda, \quad k^{FI} < k^{IM}. \quad (\text{P11.1})$$

Missing foundational step. Derive excess supply and the single-crossing condition from the household and invariant-law results rather than assume them.

Lemma 12.1: netSaving_eq_zero_of_goods_clear

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

From goods clearing $C + \delta k = F(k)$, subtraction yields $F(k) - C - \delta k = 0$.

Property 12: property12_saving_rate_sign

AMBER – Formally proved but incomplete. **Lean status:** kernel-checked, no `sorryAx`.
 If $P.s$ is strictly increasing, the capital ranking in (P11.1) gives $P.s(k^{FI}) < P.s(k^{IM})$. Lemma 12.1 applied to (E6) and (E7) makes net saving zero in both equilibria.

Missing foundational step. Prove the hypotheses of Property 11 and the monotonicity of the saving-rate formula from the production specification.

Property 13: property13_equilibrium_need_not_be_unique

RED – Intended claim not yet established. **Lean status:** kernel-checked, no `sorryAx`.
 Use one state with weight one, $\phi(r) = 1$, $A(r) = r^2$, and $K(r) = 0$. Then (P8.1) is $E_a(r) = r^2 - 1$, which is nonmonotone, and both $r = -1$ and $r = 1$ satisfy $K(r) = E_a(r) = 0$.

Missing foundational step. Certify the policy as optimal, the weights as invariant, and the capital-demand curve as arising from admissible production. The current witness is not an Aiyagari equilibrium.

Lemma 14.1: marketClearing_exists_of_sign_change

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.
 If $X = K - E_a$ is continuous on $[r_L, r_H]$, with $X(r_L) \geq 0 \geq X(r_H)$, the intermediate value theorem supplies $r \in [r_L, r_H]$ with $X(r) = 0$, equivalently (D31).

Property 14: property14_existence_under_alternative_debt_limits

RED – Intended claim not yet established. **Lean status:** kernel-checked, no `sorryAx`.
 Apply Lemma 14.1 to (K_{PV}, E_{PV}) on a positive interval and to (K_f, E_f) on a negative interval. The endpoint restrictions place the first root above zero and the second below zero.

Missing foundational step. Define these curves from the present-value and fixed-debt household models and prove the endpoint signs. As stated they are arbitrary curves, so the theorem does not establish existence under the named debt regimes.

Lemma 15.1: assetSupply_eq_capitalDemand_at_equilibrium

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.
 Market clearing is $K(r) = E_a(r)$ by (D31); symmetry gives $E_a(r) = K(r)$.

Property 15: property15_general_equilibrium_disciplines_saving

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.
 If $K(r_p) < E_a(r_p)$, then equality cannot hold at r_p , so that partial return is not an equilibrium. At a clearing return r_e , Lemma 15.1 gives $E_a(r_e) = K(r_e)$.

Property 16: property16_more_borrowing_reduces_capital

AMBER – Formally proved but incomplete. **Lean status:** kernel-checked, no `sorryAx`.
 Let E_a^0 and E_a^1 be the asset-supply schedules under the tight and relaxed limits. Lean now assumes only the economically relevant local comparison $E_a^1(r_0) < E_a^0(r_0)$ at the old clearing return, rather than an artificial constant translation of the entire schedule. Since $K(r_0) = E_a^0(r_0)$, new excess

demand at r_0 is strictly positive; at the new root it is zero. If new excess demand is strictly decreasing, this forces $r_0 < r_1$. Strictly decreasing capital demand then gives $K(r_1) < K(r_0)$.

Missing foundational step. Derive $E_a^1(r_0) < E_a^0(r_0)$ from the two solved policies and their respective invariant laws under a relaxed borrowing constraint.

Property 16b: `property16_more_borrowing_reduces_saving_rate`

AMBER – Formally proved but incomplete. **Lean status:** kernel-checked, no `sorryAx`. If $r_0 < r_1$, decreasing primitive capital demand gives $K(r_1) < K(r_0)$; applying the actual technology saving-rate function $s_P(k) = \delta k / f(k)$ preserves the inequality when s_P is strictly increasing.

Missing foundational step. Connect the assumed return comparison to the borrowing-limit model and derive saving-rate monotonicity from production.

Lemma 17.1: `effectiveBorrowingLimit_eq_natural_of_slack`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`. For $r > 0$, (D3) is the minimum of b and ϕ^{nat} . Slackness (D36) says $\phi^{nat} \leq b$, so the minimum is ϕ^{nat} .

Solved-household equality lemmas: `effectiveBorrowingLimit_admissible`, `solvedBoundedHousehold_behavior_eq_of_limit_eq`, `solvedNextExcess_eq_of_limit_eq`, `solvedResourceKernel_eq_of_limit_eq`

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`. For $r > 0$, $\phi^{eff} \leq \phi^{nat}$ implies $z_{\min} = wl_{\min} - r\phi^{eff} \geq 0$, so the effective limit is admissible. If $\phi_1 = \phi_2$, the two concrete Bellman operators are definitionally the same. Banach’s construction, the unique optimizer, optimal consumption, and the policy-induced transition therefore agree. Applying the identical random transition to the same labor law gives the same Markov kernel. Proof witnesses for admissibility lie in `Prop` and cannot change any of these economic objects.

Property 17: `property17_no_effect_when_fixed_limit_slack`

AMBER – Formally proved but incomplete. **Lean status:** kernel-checked, no `sorryAx`. If both b_1 and b_2 are slack, Lemma 17.1 makes both effective limits equal ϕ^{nat} . The theorem now applies the preceding concrete lemmas, not an arbitrary behavior function, and proves

$$V_{b_1} = V_{b_2}, \quad A_{b_1} = A_{b_2}, \quad c_{b_1} = c_{b_2}, \quad T_{b_1} = T_{b_2}, \quad P_{b_1} = P_{b_2}. \quad (\text{P17.1})$$

Consequently a probability measure is invariant for P_{b_1} if and only if it is invariant for P_{b_2} . This establishes the fixed-price household and stationary-law content of the slackness claim.

Missing foundational step. Upgrade the result to an explicit bijection between complete stationary recursive equilibrium records with statutory limits b_1 and b_2 , including the bookkeeping change in the model’s stored b field and unchanged aggregate market-clearing equations.

4.6 Properties 18–20: growth, debt, and money

Define exact growth and normalization by

$$x_t = x_0(1 + g)^t, \quad \tilde{x}_t = \frac{x_t}{(1 + g)^t}. \quad (\text{P18.1})$$

Lemma 18.1: normalizedPath_eq

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

If $1 + g \neq 0$, substitution into (P18.1) cancels the nonzero factor and gives $\tilde{x}_t = x_0$ at every date.

Property 18: property18_balanced_growth_extension

AMBER – Formally proved but incomplete. **Lean status:** kernel-checked, no `sorryAx`.

Assume $\beta > 0$, $1 + g \neq 0$, exact growth for earnings, resources, assets, and consumption, and

$$\beta(1 + r^{IM}) < (1 + g)^\gamma. \quad (\text{P18.2})$$

Since $1 + \lambda(\beta) = 1/\beta$, division of (P18.2) by $\beta > 0$ gives $r^{IM} < r_g^{FI}$ from (D37). Lemma 18.1 makes all four normalized paths constant at their initial levels.

Missing foundational step. Derive the Euler wedge and balanced-growth paths from the growing economy rather than assume them.

Lemma 19.1: debt_budget_translation

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

For gross assets $a^{net} + d$, expand (D38):

$$R_d(w, l, r, d, a^{net} + d) - d = wl + (1 + r)a^{net}. \quad (\text{L19.1})$$

The $-rd + (1 + r)d - d$ terms cancel.

Lemma 19.2: debt_budget_iff_net_budget

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Adding d to next-period net assets and using (L19.1) shows

$$c + (a'_{net} + d) = R_d(w, l, r, d, a_{net} + d) \iff c + a'_{net} = wl + (1 + r)a_{net}.$$

Lemma 19.3: taxAdjusted_floor_iff_net_natural_floor

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

For $r > 0$, definitions (D2) and (D40) give $\underline{a}_d - d = -\phi^{nat}$. Consequently

$$\underline{a}_d \leq a^{net} + d \iff -\phi^{nat} \leq a^{net}. \quad (\text{L19.2})$$

Let gross and net paths be $a_t = a_t^{net} + d$ and $a_t^{net} = a_t - d$. Debt-adjusted feasibility combines the gross budget with $a_t \geq \underline{a}_d$; debt-free feasibility combines the net budget with $a_t^{net} \geq -\phi^{nat}$.

Lemma 19.4: debtAdjustedFeasible_iff_net**GREEN – Foundationally proved.** Lean status: kernel-checked, no sorryAx.

Apply Lemma 19.2 at every date to the budget equations and Lemma 19.3 at every date to the floors. Both implications follow componentwise, giving an equivalence of the two feasible-path predicates.

Debt Bellman equivalence: debtAdjustedBellmanObjective_grossAssetOfShifted,
debtAdjustedBellmanOperator_eq_bellmanOperator,
debtAdjustedBellmanFixedPoint_of_fixedPoint,
debtAdjustedPolicy_optimal_of_optimal, debtAdjustedResourceTransition_policy

GREEN – Foundationally proved. Lean status: kernel-checked, no sorryAx.

Let $g = g_d(\hat{a}) = \hat{a} - \phi + d$. Equations (E10)–(E12) and (D38) give

$$\begin{aligned} c_d(z, g_d(\hat{a})) &= z - \hat{a}, \\ T_d(g_d(\hat{a}), l') &= T_{M, \phi}(\hat{a}, l'), \\ Q_d(V; z, g_d(\hat{a})) &= Q_{M, \phi}(V; z, \hat{a}). \end{aligned} \tag{L19.3}$$

Moreover g_d maps $[0, z]$ bijectively onto $[d - \phi, z + d - \phi]$, with inverse $g \mapsto g + \phi - d$. Hence the two sets of objective values are equal, so their suprema are equal:

$$\mathcal{B}_d V = \mathcal{B}_{M, \phi} V. \tag{L19.4}$$

Thus every debt-free Bellman fixed point is a gross-debt fixed point, and the image under g_d of every debt-free maximizing policy maximizes the actual gross-debt objective. This proves household optimality rather than assuming it as a neutrality premise.

Complete-equilibrium equivalence: taxAdjustedToDebtFree_debtFreeToTaxAdjusted,
debtFreeToTaxAdjusted_taxAdjustedToDebtFree,
taxAdjustedAssetFloor_eq_debt_sub_natural

GREEN – Foundationally proved. Lean status: kernel-checked, no sorryAx.

From a debt-free record satisfying (E13), retain (V, A, μ) and use the gross policy $g_d \circ A$. Bellman optimality follows from (L19.3)–(L19.4), and the transition equality in (L19.3) preserves the same invariant law. Integrability is preserved by addition of the constant d , while probability of μ and (E13) give

$$\int g_d(A(z)) d\mu = \int (A(z) - \phi) d\mu + d\mu(\mathbb{R}) = 0 + d = d. \tag{L19.5}$$

This constructs a complete record satisfying (E14). Subtracting d constructs the reverse map; (L19.5) then yields zero net-asset supply. The two maps preserve the household and measure fields exactly and are inverse (proof fields are propositions and hence proof-irrelevant). Finally, for $r > 0$, direct algebra gives $\underline{a}_d = d - \phi^{nat}$, so when $\phi = \phi^{nat}$ the gross feasible interval in (E10) has precisely the tax-adjusted natural floor.

Witness 19.6: fixed_floor_not_debt_neutral_witness**GREEN – Foundationally proved.** Lean status: kernel-checked, no sorryAx.

Choose $b = 0, d_0 = 0, d_1 = 1, a^{net} = -1/2$. Then $-b = 0 \leq a^{net} + d_1 = 1/2$, whereas $0 \not\leq$

$a^{net} + d_0 = -1/2$. Thus a fixed gross floor need not be preserved by debt translation.

Property 19: property19_debt_neutrality_depends_on_constraint

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Assume the household primitives satisfy (A1), $r > 0$, and $\phi = \phi^{nat}$. The explicit map from (E13) to (E14) is bijective because the two preceding inverse identities hold. It preserves Bellman values, optimal policies, consumption, resource transitions, invariant distributions, and asset-market clearing, while (L19.2) preserves every feasible path. Direct cancellation gives $(a_t^{net} + d) - d = a_t^{net}$. Witness 19.6 shows why the conclusion fails for a fixed gross borrowing floor that is not adjusted with government debt. Thus the tax-adjusted constraint gives neutrality of the complete stationary equilibrium set, whereas a fixed constraint need not.

Lemma 20.1: sharpPrice_pos

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

If $r^\#$, $w, l_{\min} > 0$, then (D42) is a positive number divided by a positive number, hence $p^\# > 0$.

Lemma 20.2: one_div_sharpPrice_eq_naturalLimit

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

When all three factors are nonzero,

$$\frac{1}{p^\#} = \frac{wl_{\min}}{r^\#} = \phi^{nat}(w, l_{\min}, r^\#). \quad (\text{L20.1})$$

Lemma 20.3: monetary_constraint_slack_above_sharp_return

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Suppose $0 < r^\# < r$, $w, l_{\min}, p > 0$, and $p < p^\#$. Since $wl_{\min}/r$ decreases in positive r ,

$$\phi^{nat}(r) < \phi^{nat}(r^\#) = 1/p^\# < 1/p.$$

Thus the monetary fixed limit $1/p$ is slack, and Lemma 17.1 makes the effective limit equal the natural limit.

Stationary mean equality:

SolvedStationaryHousehold.meanAssets_eq_of_limit_eq_unique

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Let S_1, S_2 be the concrete records (E15) for the same household model. If $\phi_1 = \phi_2$, the solved policy and full-state kernel coincide by the equal-limit household and kernel lemmas. Hence $S_1.\mu$ is invariant for the second kernel. If the second invariant probability is unique, then $S_1.\mu = S_2.\mu$. Equality of both the integrand and measure in (E15) therefore proves

$$\mathfrak{A}(S_1) = \mathfrak{A}(S_2). \quad (\text{L20.2})$$

Lemma 20.4: monetaryMeanAssets_eq_natural_of_slack

GREEN – Foundationally proved. **Lean status:** kernel-checked, no `sorryAx`.

Let S_m be the actual monetary stationary record (E16) and S_n a natural-limit stationary record for the same model. If the monetary limit is slack and S_n 's invariant probability is unique, then the two effective limits agree. The preceding theorem—not substitution into an arbitrary schedule—gives

$$\mathfrak{A}(S_m) = \mathfrak{A}(S_n). \quad (\text{L20.3})$$

Property 20: property20_monetary_return_upper_bound

AMBER – Formally proved but incomplete. **Lean status:** kernel-checked, no `sorryAx`.

Let S_m be a solved monetary equilibrium (E16) for model M , let S_n be its natural-limit stationary comparator, and let $S^\#$ be a natural-limit zero-asset equilibrium (E17) for a model $M^\#$ with the same household environment except for the return. Assume S_n 's invariant law is unique and the two concrete policy integrals obey the pairwise return ordering

$$M^\#.r < M.r \implies \mathfrak{A}(S^\#) < \mathfrak{A}(S_n). \quad (\text{P20.1})$$

If $M.r > M^\#.r$, Lemma 20.3 makes the monetary limit slack and Lemma 20.4 gives $\mathfrak{A}(S_m) = \mathfrak{A}(S_n)$. Equations (E16)–(E17) make both sides of (P20.1) zero, an impossibility. Hence $M.r \leq M^\#.r$.

Missing foundational step. Derive the full-state uniqueness premise from Property 7's compact invariant law and derive the pairwise ordering (P20.1) from the natural-limit asset-supply results. The former arbitrary supply schedule and representation hypothesis have been eliminated.

5 Declaration coverage and adequacy summary

Every declaration below is kernel-checked and independent of `sorryAx`. “Upgrade step” is therefore about economic or analytic adequacy, never logical validity.

Lean declaration	Soundness	Tier	Reason	Upgrade step
<code>consumption_nonneg</code>	Checked	Green	Directly unfolds feasible policy and consumption.	None.
<code>market_clears</code>	Checked	Green	Directly unfolds equilibrium clearing.	None.
<code>minimumIncome_shifted_budget</code>	Checked	Green	Exact budget-shift algebra.	None.
<code>discountedShiftedAssets_antitone</code>	Checked	Green	Feasibility signs the discounted recursion.	None.
<code>tendsto_naturalDebtLimit_div_pow</code>	Checked	Green	Geometric-limit argument under $r > 0$.	None.
<code>property01_naturalDebt_iff_noPonzi</code>	Checked	Green	Canonical i.i.d. proof uses predictability, independence, and positive-probability low-income blocks.	None.
<code>property01_naturalDebt_iff_noPonzi_ae</code>	Checked	Green	Exact almost-sure alias of the canonical theorem.	None.
<code>natural_bound_does_not_imply_noPonzi_without_feasibility</code>	Checked	Green	Explicit deterministic counterexample.	None.
<code>noPonzi_does_not_imply_natural_bound_without_feasibility</code>	Checked	Green	Explicit deterministic counterexample.	None.
<code>beta_mul_one_add_timePreferenceRate</code>	Checked	Green	Exact identity from the time-preference definition.	None.
<code>canonicalLabor_map_eq</code>	Checked	Green	A product-measure coordinate has the primitive labor law.	None.

Lean declaration	Soundness	Tier	Reason	Upgrade step
measure_lowerLaborBlock	Checked	Green	Finite cylinder probability is the product of marginal probabilities.	None.
lowerLaborBlock_pos	Checked	Green	Reachability makes every factor, hence the finite product, positive.	None.
exists_compactMaximum_eq	Checked	Green	Extreme-value theorem on a nonempty compact action space.	None.
compactMaximum_le	Checked	Green	Every value is below the attained supremum.	None.
abs_compactMaximum_sub_le	Checked	Green	Attained compact maxima obey the uniform perturbation bound.	None.
compactMaximum_lipschitz	Checked	Green	The perturbation estimate is exactly one-Lipschitz continuity.	None.
continuous_compactMaximum	Checked	Green	Lipschitz maps are continuous.	None.
continuousFamilyArgmax_isMax, continuous_continuousFamilyArgmax	Checked	Green	Compactness, joint objective continuity, and optimizer uniqueness prove Berge continuity by a subsequence argument.	Specialize uniqueness to each solved household objective.
maximizeObjective_norm_sub_le	Checked	Green	Statewise compact maximization is sup-norm nonexpansive.	None.
abs_integral_le_uniform_bound	Checked	Green	Probability normalization bounds an integral by its uniform envelope.	None.
abs_integral_bcf_sub_le	Checked	Green	Probability integration is sup-norm nonexpansive.	None.
expectedContinuationBCF_norm_sub_le	Checked	Green	Continuous Markov continuation preserves bounded continuity and is nonexpansive.	None.
continuous_boundedContinuousFamily, continuous_compactParameterizedExpectation_ fixedValue, continuous_ compactParameterizedExpectation	Checked	Green	Heine–Cantor and probability nonexpansiveness give sup-norm continuity on a common compact state space.	Establish a common state for the global household family.
continuousBellmanOperator_norm_sub_le	Checked	Green	Reward cancels and discounting gives the beta bound.	None.
continuousBellmanOperator_contracting	Checked	Green	The preceding estimate and beta below one prove contraction.	None.
parameterizedContractionFixedPoint_ isFixedPt, dist_ parameterizedContractionFixedPoint_ le, continuous_ parameterizedContractionFixedPoint	Checked	Green	The residual estimate (L2.5a) proves continuity of fixed points for jointly continuous uniformly contracting families.	Prove joint continuity of the concrete (b, w, r) -Bellman family.
continuous_ continuousBellmanOperator_reward_ value, continuous_ boundedBellmanValue_reward	Checked	Green	The concrete bounded Bellman map is jointly continuous in reward and candidate value, so its fixed point is continuous in reward.	Extend the operator calculation to price-dependent rewards and transitions.

Lean declaration	Soundness	Tier	Reason	Upgrade step
<code>continuous_</code> <code>compactParameterizedBellmanOperator,</code> <code>continuous_</code> <code>compactParameterizedBellmanValue,</code> <code>continuous_</code> <code>compactParameterizedSolvedObjective,</code> <code>continuous_</code> <code>compactParameterizedSolvedObjective_</code> <code>apply, continuous_</code> <code>compactParameterizedSolvedPolicy</code> <code>boundedBellmanValue_isFixedPoint</code>	Checked	Green	On common compact spaces, primitive reward/transition continuity implies solved value, objective, and unique-policy continuity.	Prove global truncation equivalence or use a weighted space.
<code>boundedBellmanValue_unique</code>	Checked	Green	Generic Banach fixed-point construction.	None.
<code>shareConsumption_nonneg</code>	Checked	Green	Generic Banach uniqueness.	None.
<code>continuous_shareNextExcess</code>	Checked	Green	Nonnegative resources and saving share imply nonnegative consumption.	None.
<code>boundedAiyagariValue_isFixedPoint</code>	Checked	Green	Arithmetic and nonnegative clipping preserve continuity.	None.
<code>boundedAiyagariValue_unique</code>	Checked	Green	Banach constructs the bounded continuous fixed point.	None.
<code>boundedAiyagariPolicy_isMax</code>	Checked	Green	Banach fixed-point uniqueness.	None.
<code>boundedAiyagariValue_concave</code>	Checked	Green	Compactness and continuity produce a pointwise maximizer.	None.
<code>boundedOptimalAsset_unique</code>	Checked	Green	Bellman iteration and closedness of concavity pass concavity to the fixed point.	None.
<code>continuous_</code> <code>boundedAiyagariAssetPolicy</code>	Checked	Green	Primitive strict utility concavity and continuation concavity give a unique optimizer.	None.
<code>hasDerivAt_of_twoSided_slope_sandwich</code>	Checked	Green	Compact-subsequence maximum-theorem proof with optimizer uniqueness.	None.
<code>hasDerivAt_of_concave_local_lowerSupport</code>	Checked	Green	Two-sided optimality slope bounds imply the envelope derivative by squeezing.	Apply it to the solved Bellman objective.
<code>LocalDominatedDerivative.hasDerivAt_integral</code>	Checked	Green	A touching differentiable lower support pins down both one-sided derivatives of a concave value.	None.
<code>HasDerivAt.nonpos_of_eventually_right_max</code>	Checked	Green	The expectation derivative follows from the explicit domination data in (A13).	None.
<code>boundedAiyagariValue_eq_optimalAssetObjective</code>	Checked	Green	Right-feasible optimality makes the derivative nonpositive.	None.
<code>boundedAiyagari_envelope_of_consumption_pos</code>	Checked	Green	The fixed point equals its objective at the constructed optimizer.	None.
<code>boundedAssetContinuation_hasDerivAt</code>	Checked	Green	Concavity and a feasible fixed-action support prove (P2.1).	None.
<code>boundedAssetObjective_hasDerivAt_at_policy</code>	Checked	Green	Dominated differentiation derives the continuation derivative.	None.
<code>boundedAiyagari_euler_inequality</code>	Checked	Green	Chain, sum, and parametric-integral rules derive the objective derivative.	None.
	Checked	Green	Boundary and interior first-order conditions prove (P2.2).	None.

Lean declaration	Soundness	Tier	Reason	Upgrade step
property02_value_and_policy_regularity	Checked	Amber	The constructed solution now includes the envelope and conditional Euler system in addition to existence, uniqueness, concavity, optimality, and state continuity.	Establish concrete Bellman/operator and optimizer parameter continuity, global domination, and weighted Inada theory.
boundedAiyagariAssetPolicy_zero_at_minimum	Checked	Green	The Euler contradiction and impatience prove zero saving at the minimum state.	None, conditional on the explicitly stated analytic closure conditions.
boundedAiyagari_marginal_le_minimum_of_policy_pos	Checked	Green	Interior Euler equality, policy monotonicity, and decreasing marginal utility prove (L3.1).	None, conditional on the explicitly stated analytic closure conditions.
boundedAiyagari_binding_cutoff	Checked	Green	Continuity and the strict impatience gap turn the marginal bound into a positive binding interval.	None, conditional on the explicitly stated analytic closure conditions.
property03_binding_region_at_low_resources	Checked	Amber	A nondegenerate binding interval is now derived from the solved policy and Euler system.	Derive positive minimum consumption and global dominated differentiation from primitives.
property03_inada_exception	Checked	Red	Positivity is assumed; the Inada condition is unused.	Derive interiority from Inada utility, feasibility, and optimality.
monotone_boundedAiyagariAssetPolicy	Checked	Green	Revealed preference and strict utility concavity prove asset monotonicity.	None.
monotone_boundedAiyagariConsumption	Checked	Green	Strict optimality and continuation concavity prove consumption monotonicity.	None.
boundedAiyagariAssetPolicy_increment_bounds	Checked	Green	The two monotonicity results imply global finite-difference bounds between zero and one.	None.
property04_monotonicity_and_one_for_one_response	Checked	Amber	Concrete weak monotonicity, continuity, and weak slope bounds are proved.	Derive strict derivatives and combine with the nondegenerate cutoff.
consumptionShiftedDown_of_policyShiftedUp	Checked	Green	Pointwise subtraction reverses the policy shift.	None.
integral_mono_of_convexOrder_concave	Checked	Green	Convex order reverses expectation order for integrable concave tests.	None.
continuationSavingGap_mono_of_meanPreservingSpread	Checked	Green	Miller-class shifted differences and convex order prove (L5.2).	None.
optimalConsumption_antitone_of_meanPreservingSpread_sameContinuation	Checked	Green	Optimality, uniqueness, and (L5.2) prove the one-step policy comparison.	None.
property05_mean_preserving_spread	Checked	Amber	Genuine convex order now yields lower consumption and higher saving for a common Miller-class continuation value.	Prove Bellman preservation and monotonicity of the difference between the two law-specific fixed points.
EventualRelativeRiskAversionBound.marginal_ratio_le_rpow	Checked	Green	Integrating the primitive differential RRA bound proves (L6.1).	None.
EventualRelativeRiskAversionBound.tendsto_marginal_ratio_add_const	Checked	Green	Marginal monotonicity, (L6.1), and squeezing prove (L6.2).	None.
rra_marginal_ratio_bounded_gap_tendsto_one	Checked	Green	The bounded labor gap and (L6.2) prove (L6.3).	None.
bounded_resource_dynamics_of_policy_tendsto_atTop	Checked	Green	Euler, envelope, RRA, and value concavity prove drift in the unbounded-policy branch.	None, conditional on labeled closures (A13)–(A14).

Lean declaration	Soundness	Tier	Reason	Upgrade step
<code>property06_bounded_resource_dynamics</code>	Checked	Amber	The solved-policy drift theorem is proved by the bounded/unbounded policy dichotomy.	Derive high-resource consumption divergence (A14) from primitives.
<code>feller_solvedResourceTransition</code>	Checked	Green	Joint continuity and compact-support parametric integration prove Feller continuity.	None.
<code>monotone_solvedResourceTransition</code>	Checked	Green	Policy monotonicity and positive gross return preserve the state order.	None.
<code>exists_invariantProbability_of_feller_compact</code>	Checked	Green	Concrete Cesàro averages, compactness, Feller continuity, and the telescoping residual prove invariant-law existence.	None.
<code>solvedHousehold_invariantProbability_exists</code>	Checked	Green	Property 6 restricts the solved Feller household to a compact absorbing interval, where Krylov–Bogoliubov applies.	None, conditional on the labeled Property 6 closures.
<code>exists_unique_invariant_and_tendsto_of_feller_mixing</code>	Checked	Green	Oscillation contraction yields geometric test-function convergence and uniqueness.	None, conditional on the explicit local kernel condition (D53).
<code>tendsto_bcfOscillation_iterate_of_finiteStepMixing</code>	Checked	Green	Complete contracting blocks dominate the finitely many remainder steps, proving (L7.9).	None.
<code>exists_unique_invariant_and_tendsto_of_feller_finiteStepMixing</code>	Checked	Green	Finite-step contraction yields uniqueness and convergence at every date of the original chain.	None, conditional on (D57).
<code>kernelOscillationMixing_of_oneStepSplitting</code>	Checked	Green	A positive common-reset event proves (L7.5) with coefficient below one.	None.
<code>minimumLaborTransition_strictly_decreases</code>	Checked	Green	Euler equality, the envelope theorem, monotone consumption, and impatience prove (L7.6).	None, conditional on the labeled dominated-differentiation input.
<code>tendsto_iterate_zero_of_monotone_strictDecrease</code>	Checked	Green	Monotone convergence and continuity identify zero as the only possible iterate limit.	None.
<code>minimumLaborBlock_uniformly_enters_bindingRegion</code>	Checked	Green	The minimum-income iterates converge to zero uniformly over every bounded initial interval by monotonicity.	None.
<code>exists_positive_lowOffset_uniformly_enters_bindingRegion</code>	Checked	Green	Continuity makes strict entrance persist on an open relative neighborhood of the lower labor endpoint.	None.
<code>lowOffsetBlockTransition_le_constant</code>	Checked	Green	Coordinatewise transition monotonicity bounds every low-offset history by the constant upper-offset path.	None.
<code>lowIncomeBlock_positive_and_uniformEntrance</code>	Checked	Green	Lower-support reachability and i.i.d. block probabilities combine with (L7.8).	None.
<code>exists_oneStepSplitting_absorbingSolvedBlockTransition</code>	Checked	Green	The supported product block has positive probability and the following shock coalesces all initial states as in (L7.10).	None.

Lean declaration	Soundness	Tier	Reason	Upgrade step
kernelIterateOscillationMixing_of_blockSplitting	Checked	Green	A block random-map representation plus common reset proves finite-step mixing.	None, conditional on the displayed representation equality.
integral_finiteIIDBlockTransition_eq_operator_iterate	Checked	Green	Finite-product Fubini proves the exact block expectation identity (L7.11).	None.
fellerMarkovOperator_finiteIIDBlock_eq_iterate	Checked	Green	Repackages (L7.11) as equality of the block Markov operator and the one-period operator iterate.	None.
exists_oneStepSplitting_finite_absorbingSolvedBlockTransition	Checked	Green	The concrete finite product contains a positive-probability common-reset block.	None.
absorbingSolvedKernel_finiteStepMixing	Checked	Green	The finite reset block and (L7.11) derive (D57) for the actual solved household kernel.	None, conditional on the displayed local analytic inputs.
property07_unique_stable_invariant_distribution	Checked	Amber	Compact-Feller existence, concrete household mixing, uniqueness, and convergence from every initial law are all proved.	Derive positive minimum-state consumption and closures (A13)–(A14) from primitives.
continuous_integral_bcf_probabilityMeasure	Checked	Green	The decomposition (L7.13) proves joint continuity in the sup-norm integrand and weak probability law.	None.
continuous_fixedPoint_of_compact_unique	Checked	Green	Compactness, joint continuity, and pointwise fixed-point uniqueness prove continuity of the fixed-point selection.	None.
continuous_parameterizedMarkovLawEvolution_of_continuous_fellerOperator	Checked	Green	Sup-norm continuity of each parameterized Feller operator implies joint continuity of law evolution.	Prove this operator continuity for the solved household.
continuous_invariantProbability_of_jointlyContinuous_evolution	Checked	Green	Specializes the compact fixed-point argument to parameterized Markov law evolution.	Prove joint law-evolution continuity for the solved household on common parameter-local absorbing intervals.
finiteStateMeanAssets_continuous	Checked	Green	A finite weighted sum preserves continuity.	None.
continuous_nonmonotone_finiteState_supply	Checked	Red	The r^2 policy is not certified optimal.	Construct an admissible household-model witness.
property08_continuity_and_possible_nonmonotonicity	Checked	Red	It inherits the uncertified r^2 witness.	Verify policy optimality, invariance, and admissible primitives.
reciprocal_gap_diverges	Checked	Green	Standard reciprocal limit.	None.
property09_explosion_at_time_preference_rate	Checked	Amber	Explosion uses a supplied divergent lower bound.	Derive the bound from the Euler equation and stationary law.
property10_uncertainty_raises_assets_near_benchmark	Checked	Amber	Continuity propagates an assumed precautionary gap.	Prove the gap from a mean-preserving spread.
property11_factor_price_and_capital_signs	Checked	Amber	Excess-demand signs and single crossing are inputs.	Derive them from household supply and production.
netSaving_eq_zero_of_goods_clear	Checked	Green	Goods clearing yields the identity algebraically.	None.
property12_saving_rate_sign	Checked	Amber	Saving comparisons use supplied monotonicity.	Derive factor-price and saving-rate shapes from primitives.
property13_equilibrium_need_not_be_unique	Checked	Red	Arbitrary aggregation and zero capital demand are used.	Certify two market-clearing equilibria of an admissible model.

Lean declaration	Soundness	Tier	Reason	Upgrade step
marketClearing_exists_of_sign_change	Checked	Green	Correct intermediate-value theorem argument.	None.
property14_existence_under_alternative_debt_limits	Checked	Red	The two curves are not linked to the named debt regimes.	Build both household supply curves and prove endpoint signs.
assetSupply_eq_capitalDemand_at_equilibrium	Checked	Green	Direct consequence of equilibrium clearing.	None.
property15_general_equilibrium_disciplines_saving	Checked	Green	Uses earlier equilibrium identities exactly.	None.
property16_more_borrowing_reduces_capital	Checked	Amber	A local supply comparison and excess-demand monotonicity imply the equilibrium return and capital ordering.	Derive the local supply comparison from household invariant laws.
property16_more_borrowing_reduces_saving_rate	Checked	Amber	The actual technology saving-rate function preserves the capital ordering when strictly increasing.	Derive its monotonicity from production primitives.
effectiveBorrowingLimit_eq_natural_of_slack	Checked	Green	Minimum of two ordered limits is the natural limit.	None.
effectiveBorrowingLimit_admissible, solvedBoundedHousehold_behavior_eq_of_limit_eq, solvedNextExcess_eq_of_limit_eq, solvedResourceKernel_eq_of_limit_eq	Checked	Green	Equal effective limits give identical solved values, policies, consumption, transitions, and kernels.	None.
property17_no_effect_when_fixed_limit_slack	Checked	Amber	Slack statutory limits now produce identical concrete household behavior, kernels, and invariant-law sets.	Build the corresponding bijection of complete equilibrium records.
normalizedPath_eq	Checked	Green	Exact growth factors cancel.	None.
property18_balanced_growth_extension	Checked	Amber	Euler wedge and growth paths are assumed.	Derive them from the growing household economy.
debt_budget_translation	Checked	Green	Exact cancellation in the debt budget.	None.
debt_budget_iff_net_budget	Checked	Green	Algebraic equivalence of gross and net budgets.	None.
taxAdjusted_floor_iff_net_natural_floor	Checked	Green	Exact translation of the borrowing floor.	None.
debtAdjustedFeasible_iff_net	Checked	Green	Pointwise budget-and-floor equivalence.	None.
debtAdjustedBellmanObjective_grossAssetOfShifted, debtAdjustedBellmanOperator_eq_bellmanOperator, debtAdjustedBellmanFixedPoint_of_fixedPoint, debtAdjustedPolicy_optimal_of_optimal, debtAdjustedResourceTransition_policy	Checked	Green	The affine coordinate change preserves the actual gross-debt objective, Bellman operator, fixed point, maximizing policy, and resource transition.	None.
taxAdjustedToDebtFree_debtFreeToTaxAdjusted, debtFreeToTaxAdjusted_taxAdjustedToDebtFree, taxAdjustedAssetFloor_eq_debt_sub_natural	Checked	Green	The complete stationary-equilibrium translations are inverse and the gross floor is exactly debt minus the natural limit.	None.
fixed_floor_not_debt_neutral_witness	Checked	Green	Explicit fixed-floor counterexample.	None.
property19_debt_neutrality_depends_on_constraint	Checked	Green	Proves a bijection of complete debt-free and tax-adjusted debt stationary equilibrium sets, plus path equivalence and the fixed-floor counterexample.	None.
sharpPrice_pos	Checked	Green	Quotient of positive terms.	None.

Lean declaration	Soundness	Tier	Reason	Upgrade step
<code>one_div_sharpPrice_eq_naturalLimit</code>	Checked	Green	Exact reciprocal identity.	None.
<code>monetary_constraint_slack_above_sharp_return</code>	Checked	Green	Monotonic natural-limit comparison.	None.
<code>SolvedStationaryHousehold.meanAssets_eq_of_limit_eq_unique</code>	Checked	Green	Equal solved limits and invariant-law uniqueness identify the actual policy integrals.	None for its stated theorem.
<code>monetaryMeanAssets_eq_natural_of_slack</code>	Checked	Green	Slackness, solved-kernel equality, and uniqueness identify monetary and natural-limit mean assets.	None for its stated theorem.
<code>property20_monetary_return_upper_bound</code>	Checked	Amber	Uses concrete solved policies, invariant laws, and market-clearing zeros; only full-state uniqueness and the cross-return integral ordering remain inputs.	Derive uniqueness from Property 7 and the ordering from natural-limit asset supply.

Interpretation. Green, amber, and red do not grade Lean’s logical kernel: all entries are valid theorems of their displayed hypotheses. They grade whether those hypotheses and conclusions constitute the foundational economic proof suggested by the declaration’s role in the Aiyagari formalization.